DEVELOPMENT OF LOW-COST GATEWAY FOR LORAWAN

A project report submitted in partial fulfillment of the requirements for the award of the Degree

of

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

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Accredited by NAAC with "A+" grade Cherukupalli (V),Near Tagarapuvalasa Bridge,

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This is to certify that the project work entitled "**DEVELOPMENT OF LOW-COST GATEWAY FOR LORAWAN"** is a Bonafide work carried out by **BANTUPALLI SRI KAVYA (23Q75A0404),** submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in **Electronics and Communication Engineering**, AVANTHI INSTITUTE OF ENGINEERNG AND TECHNOLOGY Vizianagaram during the academic year 2022-2026

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DECLARATION

We hereby declare that the project work entitled "**DEVELOPMENT OF LOW-COST GATEWAY FOR LORAWAN"** is an original work done in the Department of Electronics and Communication Engineering, AVANTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY submitted in partial fulfillment of the requirements for the award of the degree of **B.Tech. in Electronics and Communication Engineering**. The work has not been submitted to any other college or university for the award of any degree or diploma.

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ABSTRACT

The project aims to develop an affordable LoRaWAN gateway using Arduino, enabling seamless communication between sensor-equipped nodes and cloud platforms. LoRaWAN technology provides an efficient and long-range wireless solution, ideal for connecting IoT devices in remote or challenging environments. By utilizing the Arduino as the gateway, we leverage its cost-effectiveness and versatility, making it accessible for a wide range of applications.

In this project, nodes are equipped with various sensors that can securely transmit data to the Arduino gateway. The gateway acts as an intermediary, collecting data from multiple nodes and facilitating their transmission to the cloud.

The Arduino gateway is equipped with LoRaWAN. Additionally, the gateway is designed to support multiple communication channels, allowing compatibility with various cloud platforms.

This project will therefore replace conventional gateways with a low-cost gateway and can be used in various applications in daily life like in agriculture, aquaculture, Industries etc.

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CHAPTER – I
LoRa AND LoRaWAN

CHAPTER – 1 LoRa AND LoRaWAN

1.1 Introduction to LoRa and LoRaWAN

LoRa is a wireless modulation technique that stands for "long range". It's a spread spectrum modulation technique derived from chirp spread spectrum (CSS) technology. LoRa encodes information on radio waves using chirp pulses. LoRa modulated transmission is robust against disturbances and can be received across great distances. LoRa is widely acknowledged for its ability to send small amounts of data over wide distances. LoRaWAN data transfer speeds range from 250 bits per second (bps) in the longest-range setting to roughly 22 kilobits per second (Kbps) in its fastest shortest-range configuration. At the longest ranges, it can take several seconds to transmit a 1 KB message. LoRa is based on chirp spread spectrum modulation, which has low power characteristics like FSK modulation but can be used for long range communications. LoRa can be used to connect sensors, gateways, machines, devices, animals, people etc. wirelessly to the cloud.

Fig. 1.1: LoRa.

LoRa Technologies operates in different frequency bands in different regions: In the United States it operates in the 915 MHz band, in Europe it operates in the 868 MHz band and in Asia it operates in the 865 to 867 MHz, 920 to 923 MHz band.

LoRaWAN is a Low Power, Wide Area (LPWA) networking protocol developed by the LoRa Alliance, that wirelessly connects battery operated 'things' to the internet in regional, national or global networks, targeting key Internet of Things (IoT) requirements such as bi-directional communication, end-to-end security, mobility and localization services.

LoRaWAN uses unlicensed spectrum in the ISM bands to define the communication protocol and system architecture for the network while the LoRa physical layer creates the long-range communication links between remote sensors and gateways connected to the network. This protocol helps in the quick setup of public or private IoT networks anywhere using hardware and software. LoRaWAN is a Media Access Control (MAC) layer protocol built on top of LoRa modulation. It is a software layer which defines how devices use the LoRa hardware, for example when they transmit, and the format of messages.

LoRaWAN is suitable for transmitting small size payloads (like sensor data) over long distances. LoRa modulation provides a significantly greater communication range with low bandwidths than other competing wireless data transmission technologies. The following figure shows some access technologies that can be used for wireless data transmission and their expected transmission ranges vs. bandwidth.

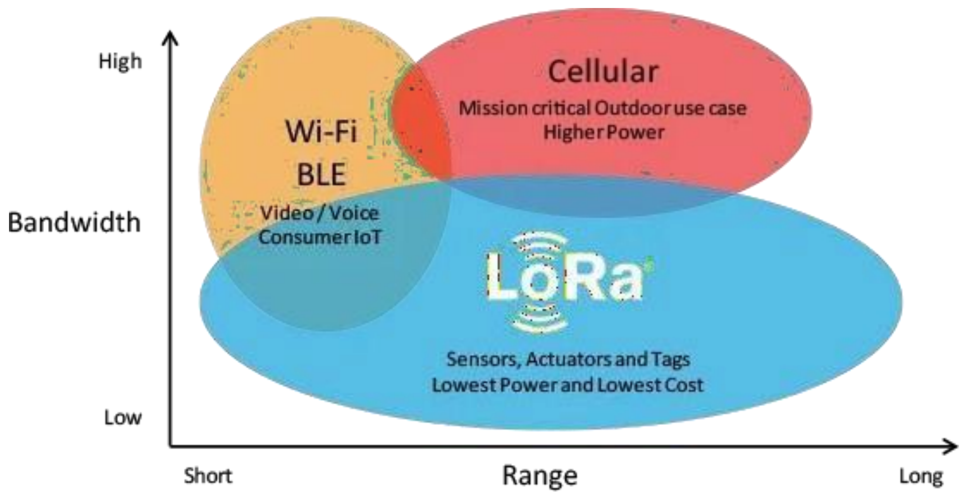

Fig. 1.2: LoRaWAN Bandwidth Vs Range Curve.

1.2 History of LoRa

The story of LoRa began in 2009 when two friends Nicolas Sornin and Olivier Seller from France, aimed at developing a long-range, low power modulation technology. In 2010 François Sforza joined the team, and together they started the company Cycleo. Initially, the three founders targeted the metering industry and aimed at adding wireless communication capabilities for gas, water, and electricity meters. For this purpose, they used Chirp Spread Spectrum (CSS) modulation technology.

Later, they Convinced Semtech about the invention's long-range and low power capabilities; Semtech acquired Cycleo in May 2012. Finally, Semtech collaborated with Nicolas, Olivier, and François to improve the technology and finalize the chips required for the end devices (SX1272 and SX1276) and the gateways (SX1301).

Fig. 1.3: Inventors of LoRa.

1.3 Chirp Modulation in LoRa

Chirp modulation is a type of modulation that uses sinusoidal waveforms whose frequency increases or decreases linearly over time. These waveforms are commonly referred to as linear chirps or simply chirps. The rate at which their frequency changes is called the chirp rate. Chirp modulation is more advanced, and closer to the modulation that your Wi-Fi uses. But also, it is not quite the same as Wi-Fi. That family is called digital modulation, and more specifically, spread-spectrum modulation.

LoRa modulation is based on chirp spread spectrum technology, allowing for long-range communication while maintaining low power consumption. It operates in the sub-gigahertz ISM bands, providing robust communication even in challenging environments.

Spread spectrum modulation is a modulation technique that was originally used in military applications to avoid detection of wireless communication by hiding it in noise but has since found its way into the consumer space.

In exact terms, a chirp is a signal in which frequency increases or decreases with time. Let's break down what that means by comparing it to the less complex, but better-known modulation techniques like AM and FM. FM varies the signal frequency on top of a carrier frequency. You might be familiar with FM radio broadcast, where carrier is in the MHz range, while the modulated signal is 54 kHz for stereo signal, 59 kHz with RDS. AM and FM are both analog modulations, designed to carry an analog signal (such as audio). We need to make quite a leap to the chirp modulation, which is not an analog but a digital modulation.

Chirp modulation takes the 1s and 0s of the data you want to send and converts them into a set of chirps. The chirps are sent one after the other on top of a fixed-frequency carrier signal. Unfortunately, the chirp modulated signal is too complex to illustrate in a simple diagram.

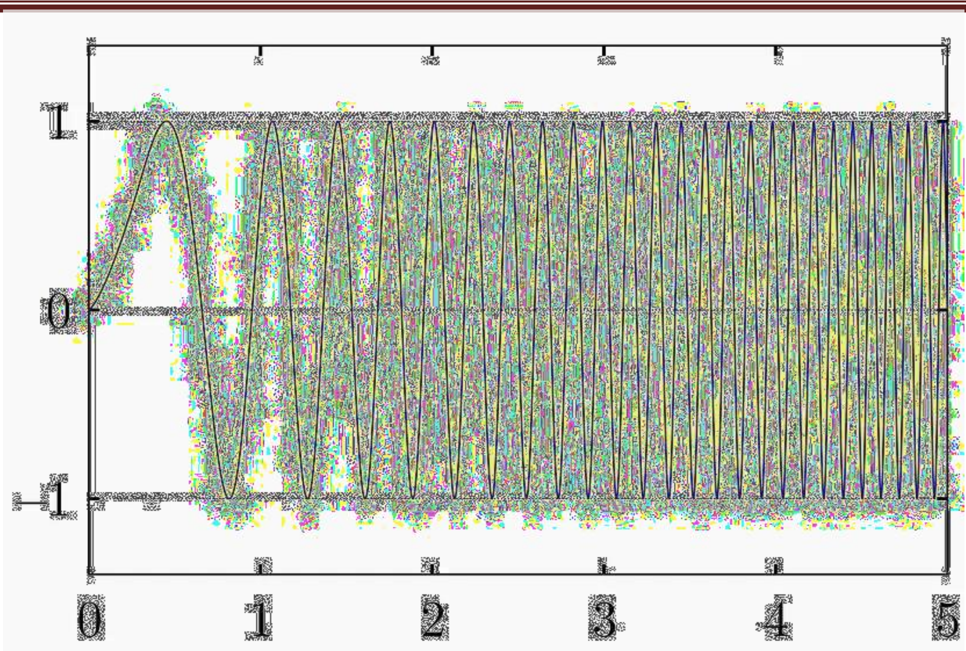

Fig. 1.4: Chirp Modulation

- **LoRa Bandwidth:** Bandwidth is the rate at which a signal changes and defines the theoretical maximum of information that can be communicated per unit of time. The bandwidth unit is Hertz (Hz). In theory, you could use any bandwidth with LoRa. In practice, you will always be limited by the available spectrum, spectrum regulations, and physical limits of your transceiver. Because of that, the commonly used bandwidth (channel width) in sub-GHz LoRa is 125 kHz, while most chips can also modulate at 250 kHz and 500 kHz, and lower frequencies too. In 2.4 GHz, LoRa chips can handle up to 1.625 MHz of bandwidth. The rule of thumb for bandwidth is
  - The higher the bandwidth, the higher the data rate (chip rate)
  - The lower the bandwidth, the lower the data rate (chip rate)
- **Lora Spreading Factor:** A spreading factor defines the ratio between the bandwidth (chip rate) and the data rate (symbol rate). It defines how much a symbol should be spread over time and frequency. Basically, it says how quick or how slow the chirp sweep should be over the selected bandwidth.

Spreading Factor (For UL at 125 KHz)	Bit Rate	Range (Depends on Terrain)	Time on Air for an 11-byte payload
SF10	980 bps	8 km	371 ms
SF9	1760 bps	6 km	185 ms
SF8	3125 bps	4 km	103 ms
SF7	5470 bps	2 km	61 ms

Table. 1.1: LoRa Spreading Factor.

- The higher the spreading factor, the more spectrum it eats, but the easier it is for the other side to decode the signal. That means you can use the spreading factor to increase the range at which a LoRa signal can still be successfully demodulated. No need for a bigger antenna, or a more powerful amplifier. It's all just configuration.
- Eating spectrum means that if your bandwidth is fixed (which is the case in LoRaWAN networks, and most LoRa networks), the higher the spreading factor you use, the lower the data rate (symbol rate) you get. The higher the spreading factor, the longer the Time on Air (ToA) the longer your transmission will take.
- This is where the real power of LoRa is. You can decide, in software, whether you need more range, more bandwidth, or shorter transmissions (lower battery use).

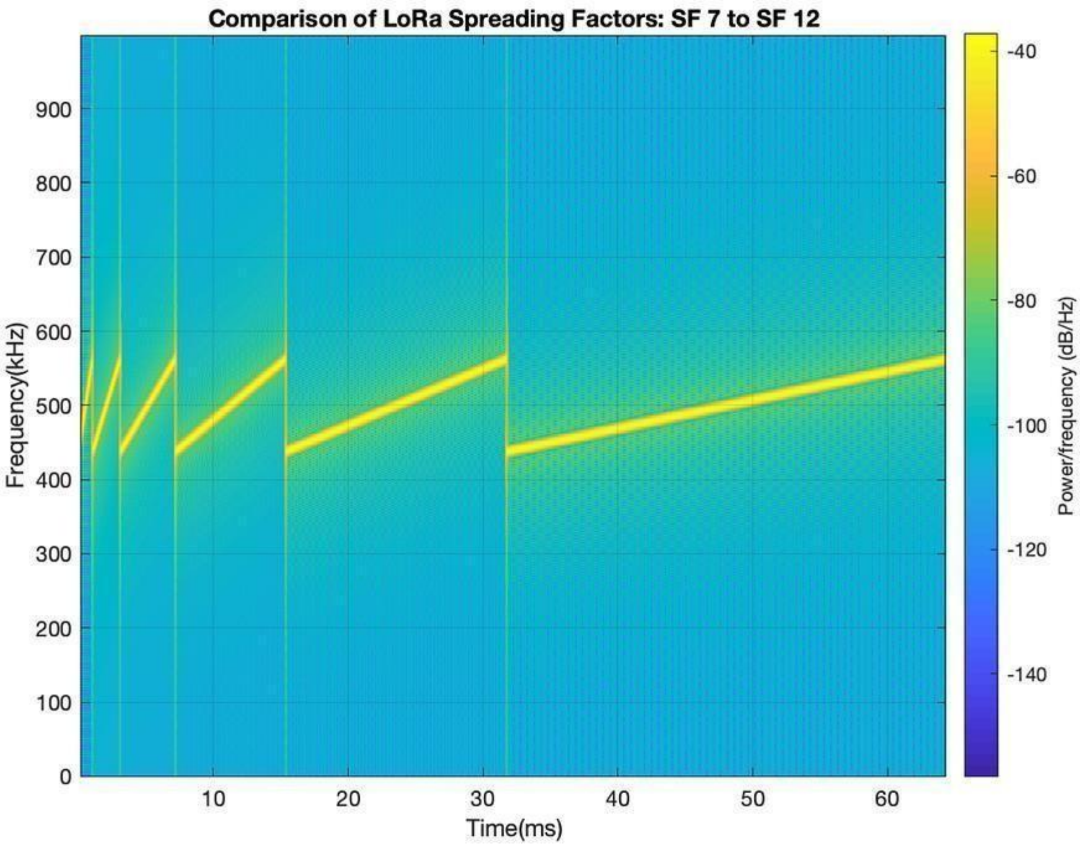

Fig. 1.5: Comparison of LoRa Spreading Factor: SF7 to SF12.

- In theory, you could use any spreading factor with LoRa. In practice, you are limited by the available bandwidth and by the demodulator design. The physical microchip size, microchip complexity, and other commercial constraints will define what LoRa modulators and demodulators will be capable of holding the data.
- While the original LoRa chips had a spreading factor range from SF7 to SF12, modern modulators in modern chips have a range from SF5 to SF12.

SYMBOL RATE in chips/s = BANDWIDTH in Hz divided by (2 to the power of SPREADING FACTOR)

- Spreading factor TL, DR:
    - The higher the spreading factor, the lower the data rate.
    - The higher the spreading factor, the longer the range.
    - The higher the spreading factor, the slower the transmission will be.
    - The lower the spreading factor, the higher the data rate.
    - The lower the spreading factor, the shorter the range.
    - The lower the spreading factor, the faster the transmission will be.

1.4 LoRaWAN Network Architecture

To fully understand LoRaWAN networks, we will start with a look at the technology stack. As shown in, LoRa is the physical (PHY) layer, i.e., the wireless modulation used to create the long-range communication link. LoRaWAN is an open networking protocol that delivers secure bi-directional communication, mobility, and localization services standardized and maintained by the LoRa Alliance.

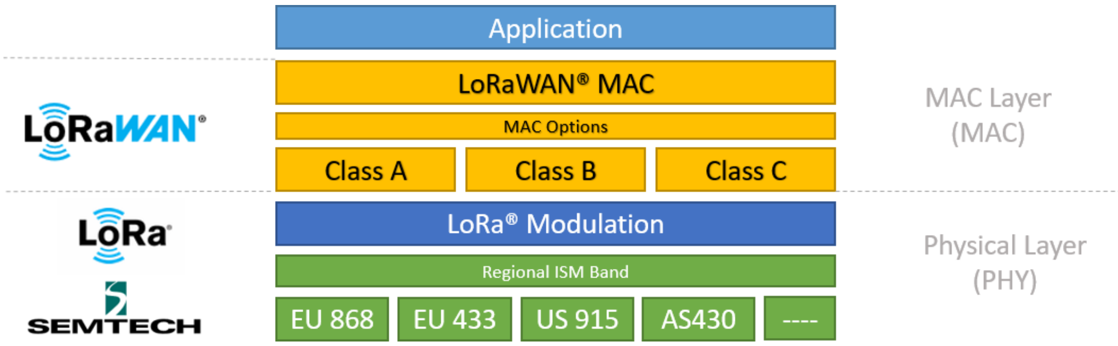

Fig. 1.6: LoRaWAN Technology stack.

The capacity of a LoRaWAN network is a function of its gateway density. To maximize the capacity of the network, using an adaptive data rate (ADR) mechanism is essential. The main goal of ADR is to save the battery power of the LoRaWAN end-nodes. By having the end-nodes closest to a gateway transmit using the lowest spreading factor, their time on air is minimized, thereby prolonging their battery life. More distant sensors transmit at a higher spreading factor. A trade-off is made between battery power and distance given that a higher spreading factor allows for a gateway to connect to devices that are farther away.

LoRaWAN Network Elements:

- **LoRa based End Devices:** A LoRaWAN-enabled end device is a sensor or an actuator which is wirelessly connected to a LoRaWAN network through radio gateways using LoRa RF Modulation.
- In the majority of applications, an end device is an autonomous, often battery-operated sensor that digitizes physical conditions and environmental events. Typical use cases for an actuator include street lighting, wireless locks, water valve shut off, leak prevention, among others.

- When they are being manufactured, LoRa-based devices are assigned several unique identifiers. These identifiers are used to securely activate and administer the device, to ensure the safe transport of packets over a private or public network and to deliver encrypted data to the Cloud.

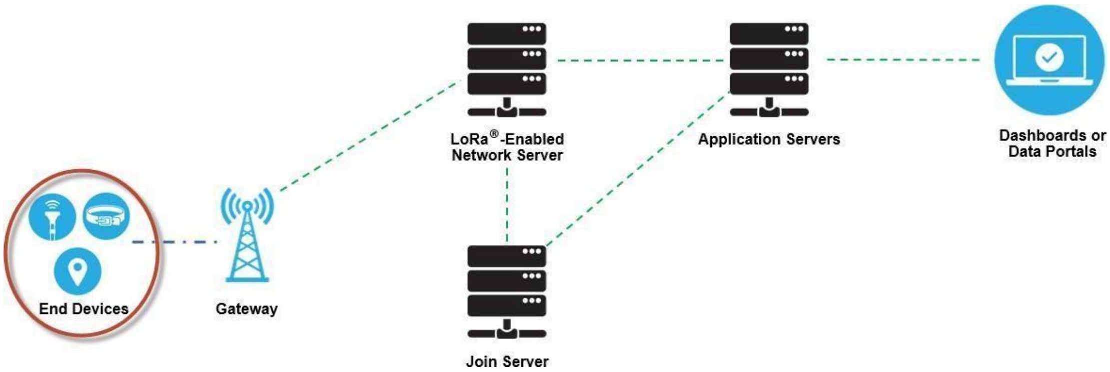

Fig. 1.7 End devices in a typical LoRaWAN Network.

- **LoRaWAN Gateways:** A LoRaWAN gateway receives LoRa modulated RF messages from any end device in hearing distance and forwards these data messages to the LoRaWAN network server (LNS), which is connected through an IP backbone. There is no fixed association between an end device and a specific gateway.
- Instead, the same sensor can be served by multiple gateways in the area. With LoRaWAN, each uplink packet sent by the end-device will be received by all gateways within reach. This arrangement significantly reduces packet error rate (since the chances that at least one gateway will receive the message are very high), significantly reduces battery overhead for mobile/nomadic sensors, and allows for low-cost geolocation (assuming the gateways in question are geolocation-capable).
- The IP traffic from the gateway to the network server can be backhauled via Wi-Fi, hardwired Ethernet or via a cellular connection. LoRaWAN gateways operate entirely at the physical layer and, in essence, are nothing but LoRa radio message forwarders. They only check the data integrity of each incoming LoRa RF message. If the integrity is not intact, that is, if the CRC is incorrect, the message will be dropped. If correct the gateway will forward it to the LNS, together with some metadata that includes the receive RSSI level of the message as well as an optional timestamp.
- For LoRaWAN downlinks, a gateway executes transmission requests coming from the LNS without any interpretation of the payload. Since multiple gateways can receive the same LoRa RF message from a single end device, the LNS performs data de-duplication and deletes all copies.
- Based on the RSSI levels of identical messages, the network server typically selects the gateway that received the message with the best RSSI when transmitting a downlink message because that gateway is the one closest to the end device in question.

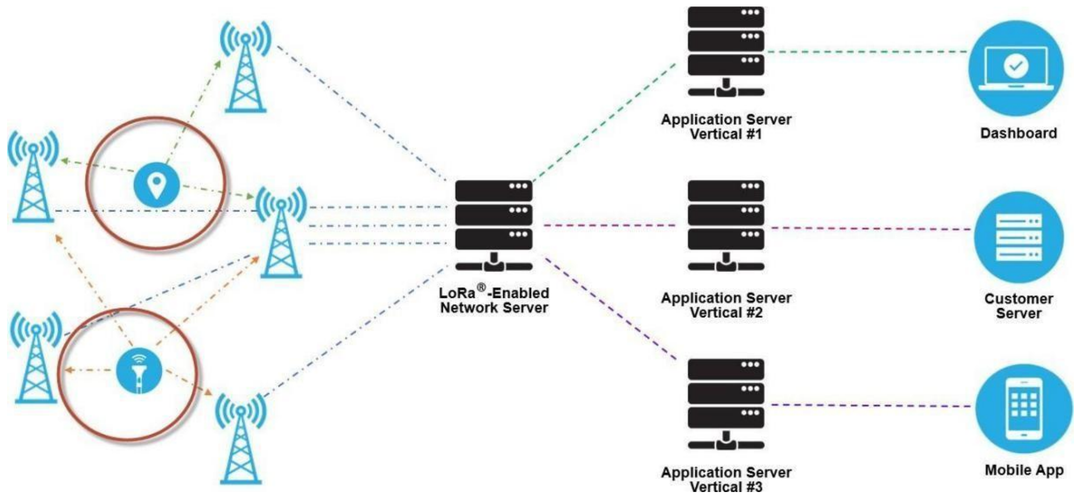

Fig. 1.8: Gateways in a typical LoRaWAN Network.

1.5 Key Features of LoRaWAN

- **Geo-Location:** It enables GPS-free, low power tracking technology
- **Low Cost:** Reduces costs three ways: infrastructure investment, operating expenses and end-node sensors, A LoRa base station cost a few hundred dollars and set themselves up with a network. It is easy to plug into the existing infrastructure and offers a solution to serve battery-operated IoT applications
- **Standardization:** LoRa Alliance develops global standards and improved global inter-operability speeds adoption and roll out of LoRaWAN-based networks and IoT applications
- **Low Power:** Its protocol is designed specifically for low power consumption extending battery lifetime up to 20 years
- **Long Range:** Single base station of LoRa provides deep penetration in dense urban/indoor regions, plus connects rural areas up to 30 miles away
- **High Capacity:** It supports millions of messages per base station, ideal for public network operators serving many customers
- **Secure:** Embedded end-to-end AES-128 encryption mechanism
- **Low connection cost:** LoRa Technology operates in the unlicensed ISM band, which means no or very low spectrum costs (there may be a very low connection fee if using an external service provider).

1.6 Comparison between LoRaWAN and other wireless technologies

When comparing LoRa with other wireless communication technologies, it's essential to consider various factors such as range, power consumption, data rate, deployment cost, and use cases. Here is a comparison between LoRa and some other popular wireless technologies:

LoRa vs. Wi-Fi:

- **Range:** LoRa offers a longer range (up to several kilometers) compared to Wi-Fi, which typically has a range of up to a few hundred meters.
- **Power Consumption:** LoRa consumes significantly less power compared to Wi-Fi, making it suitable for battery-powered IoT devices that need long-lasting operation.
- **Data Rate:** Wi-Fi provides higher data rates compared to LoRa, making it more suitable for applications that require high-speed data transfer, such as video streaming and large file transfers.
- **Deployment Cost:** LoRa deployment costs are generally lower than Wi-Fi, especially for applications requiring extensive coverage over a wide area.

LoRa vs. Cellular (3G/4G/5G):

- **Range:** LoRa provides a much longer range compared to cellular networks, making it more suitable for applications requiring communication over large distances, especially in remote and rural areas.
- **Power Consumption:** LoRa consumes significantly less power than cellular networks, allowing for extended battery life in IoT devices.
- **Data Rate:** Cellular networks offer higher data rates compared to LoRa, making them more suitable for applications that require high-speed data transfer and real-time communication.
- **Deployment Cost:** LoRa deployment costs are generally lower than those of cellular networks, especially for large-scale IoT deployments that require extensive coverage and minimal infrastructure.

LoRa vs. Bluetooth:

- **Range:** LoRa has a much longer range compared to Bluetooth, making it more suitable for applications that require communication over longer distances and in environments with obstacles.
- **Power Consumption:** LoRa consumes significantly less power than Bluetooth, allowing for extended battery life in IoT devices, especially for applications that involve low data rate and periodic communication.
- **Data Rate:** Bluetooth provides higher data rates compared to LoRa, making it more suitable for applications that require real-time data transfer, such as audio streaming and device-to-device communication.
- **Deployment Cost:** LoRa deployment costs are generally lower than Bluetooth, especially for applications that require long-range communication and deployment over a wide area.

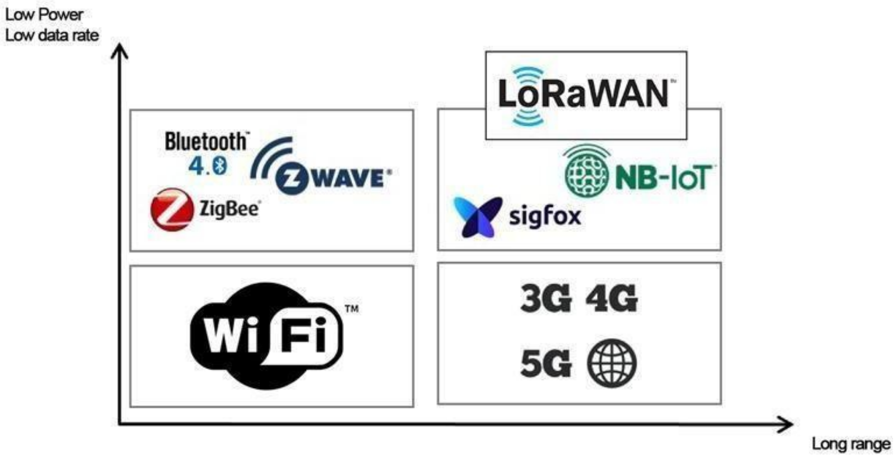

Fig. 1.9: Comparison between LoRaWAN and other wireless technologies.

1.7 Applications of LoRaWAN

LoRa and LoRaWAN have found practical applications in various industries such as:

- Air Pollution Monitoring
- Agriculture Processing
- Animal Tracking
- Fire Detection
- Fleet Tracking
- Home Security
- Indoor Air Quality
- Industrial Temperature Monitoring
- Assets Management
- Predictive Maintenance
- Radiation Leak Detection
- Smart Lighting
- Smart Parking
- Waste Management
- Water Flow Monitoring

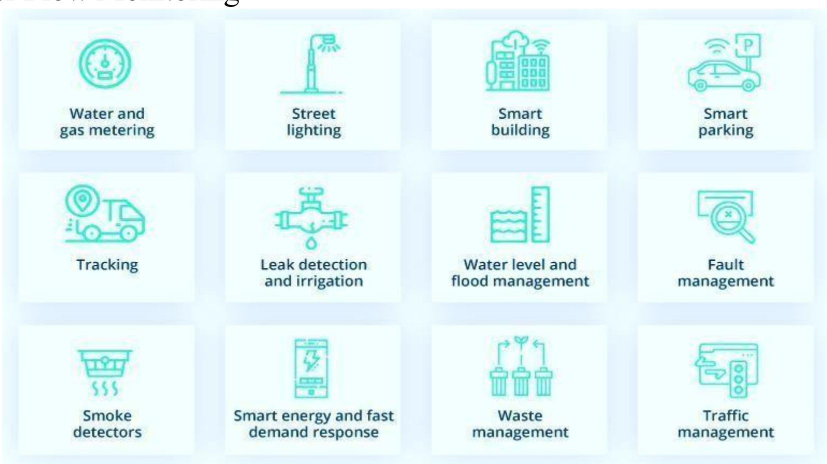

Fig. 1.10: LoRaWAN Applications.

CHAPTER – II
LoRaWAN GATEWAY

CHAPTER – 2
LoRaWAN GATEWAY

2.1 Introduction to Gateway

A gateway is a data communication device that provides a remote network with connectivity to a host network. A gateway device provides communication to a remote network or an autonomous system that is out of bounds for the host network nodes. Gateways serve as the entry and exit point of a network; all data routed inward or outward must first pass through and communicate with the gateway in order to use routing paths. Generally, a router is configured to work as a gateway device in computer networks.

Any network has a boundary or a limit, so all communication placed within that network is conducted using the devices attached to it, including switches and routers. If a network node wants to communicate with a node/network that resides outsides of that network or autonomous system, the network will require the services of a gateway, which is familiar with the routing path of other remote networks.

The gateway (or default gateway) is implemented at the boundary of a network to manage all the data communication that is routed internally or externally from that network. Besides routing packets, gateways also possess information about the host network's internal paths and the learned path of different remote networks. If a network node wants to communicate with a foreign network, it will pass the data packet to the gateway, which then routes it to the destination using the best possible path.

- Gateways make communication between the application and the server feasible by connecting them altogether through a network tube, an imaginary tube that is believed to exist between the application and the server and constituted by network waves through which data transmission takes place.
- A request about a certain amount of data is made by the user end and then the server finds and processes data accordingly with the request.
- The transmission is made via network tubes through the server to the gateway and then the gateway to the application that made a request for data.
- Gateways are considered to be the backbone for preserving intelligence as well as the storage capacity of the smart device altogether without compromising on the loss of the duo.
- Gateways do this by application of fog computing, on which a brief is provided below that helps you to get an overview of the deal between gateways and fog computing and the whole data processing thing.

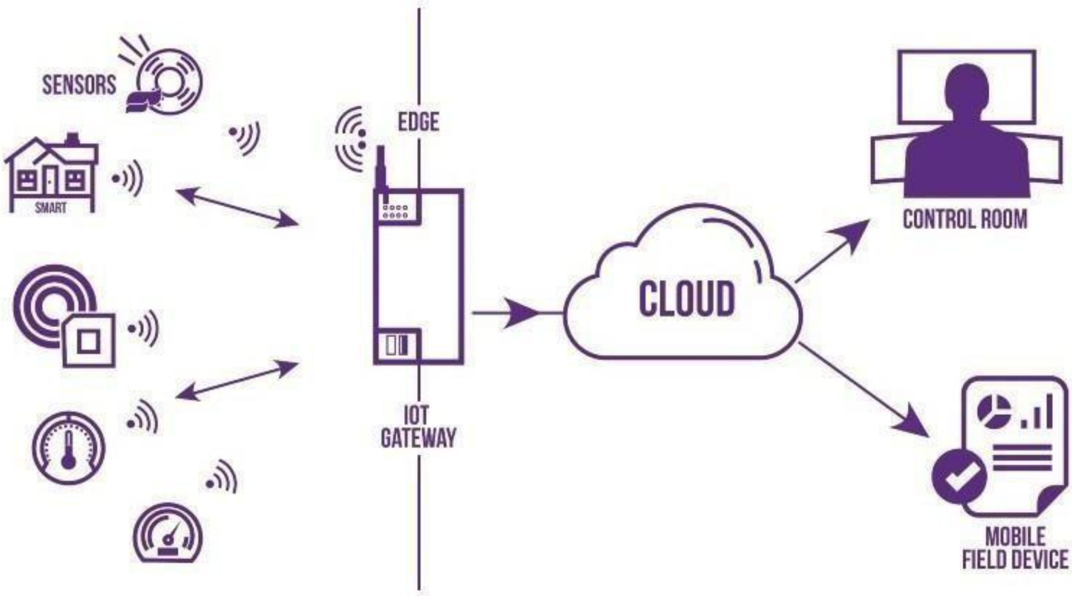

Fig. 2.1: Gateway.

2.2 Types of Gateways

There is not any specification of gateways in the commercial market other than being brand specific. But on the feasibility, performance, speed, and workability gateways can be classified in a broad manner as:

- **Unidirectional Gateways:** Data can only pass through unidirectional gateways in one direction. The destination node replicates changes made in the source node but not the other way around. They are tools for archiving.
- **Bidirectional Gateway:** Data can pass through bidirectional gateways in both directions. They are tools for synchronization.

Examples of Gateways:

- **Network Gateway:** The most popular kind of gateway, known as a network gateway, act as an interface between two disparate networks using distinct protocols. Anytime the word gateway is used without a type designation, it refers to a network gateway.
- **Cloud Storage Gateway:** A network node or server known as a cloud storage gateway translates storage requests made using various cloud storage service API calls, such as SOAP (Simple Object Access Protocol) or REST (Representational State Transfer). Data communication is made simpler since it makes it easier to integrate private cloud storage into applications without first moving those programmes to a public cloud.
- **Internet-To-Orbit Gateway (I2O):** Project HERMES and Global Educational Network for Satellite Operations (GENSO) are two well-known I2O gateways that connect devices on the Internet to satellites and spacecraft orbiting the earth.

- **IoT Gateway:** Before delivering sensor data to the cloud network, IoT gateways assimilate it from Internet of Things (IoT) devices in the field and translate between sensor protocols. They link user applications, cloud networks, and IoT devices.
- **VoIP Trunk Gateway:** By using a VoIP (voice over Internet Protocol) network, it makes data transmission between POTS (plain old telephone service) devices like landlines and fax machines easier.

2.3 Features of a Gateway

Gateways provide a wide variety of features which are:

- Gateways work as a network bridge for data transmission as it makes the transmission of data possible to transmit with more ease and does not demand high storage capacity.
- Gateways create a structural temporary storeroom for the data transmitted by the server and data requests made by the user end.
- Gateways made the transmission more feasible as it queued up all the data and divided it into small packets of data rather than sending it bulk. Data transmitted through Gateway is divided into various useful and small packets each having its individual significance and a role to play while processing data.
- Gateways made the data more secure if the modifications to the gateway could be done which then could create more reliability over smart devices.
- Gateways optimize the data for search engines, applications, and servers by implanting better readability to the content so that a machine can understand and optimize data with ease.

2.4 Comparison between Router and Gateway

The gateway is often the modem (or modem-router combo) provided by your Internet Service Provider (ISP). This gateway allows your computers and devices to connect to the ISP's network. On the ISP's end of things, the computer that controls all of the data that your ISP sends and receives is an additional node.

A router is a piece of hardware that connects you to the internet. For an in-home network, the router comes with special software that you install on one computer. Then, you can establish a home network so that anyone you allow on your network can access the ISP – and therefore the internet. Routers can be connected to two or more networks at a time, but that's not usually the case for a simple home network.

Differences Between Gateways & Routers:

There can be some confusion between gateways, routers, and other kinds of networking devices. Here is a list of the devices that are commonly used to get online.

- **Modem:** Gets its name from "modulator-demodulator;" a piece of hardware that connects to the internet.
- **Router:** A piece of hardware that connects the modem to a computer network and also makes sure that traffic is routed through the correct network.

- **Switch:** A piece of hardware that connects multiple devices within a single network; the switch transfers incoming and outgoing traffic between these devices.
- **Access Point:** A piece of networking hardware that connects devices through a WIFI connection.
- **Gateway:** A node that regulates traffic between at least two (but often more) networks that are dissimilar from one another.

A router can be a network gateway. So can a modem. If a device connects to the internet and translates information between two or more networks, then it's a gateway.

A router can control the path through which information is sent in and out. This is done by using built-in headers and forwarding tables that determine where each packet of data needs to be sent. These packets contain your emails, transactions, online activity, and more.

2.5 LoRaWAN Gateway

LoRaWAN gateway typically refers to the physical box or encasement housing the hardware and application software that performs essential tasks to connect IoT devices to the cloud. IoT devices use a gateway as a central hub to drop sensed knowledge and connect that data to external networks. A gateway is much like a Wi-Fi router. It has a LoRa concentrator, allowing it to receive RF signals sent out by LoRaWAN devices, which get converted to a signal compatible with a server, such as Wi-Fi, to send data to the cloud.

A gateway is a jumping-off point as all the devices on the network have to be able to transmit their data back to the gateway. This often means large scale LoRaWAN systems run on more than one gateway. The maximum number of sensors or devices per gateway depends on a number of factors such as data rate and data packet size but hundreds or even low thousands per gateway is achievable. Gateways are a crucial component to running efficient wireless networks for sensors and devices.

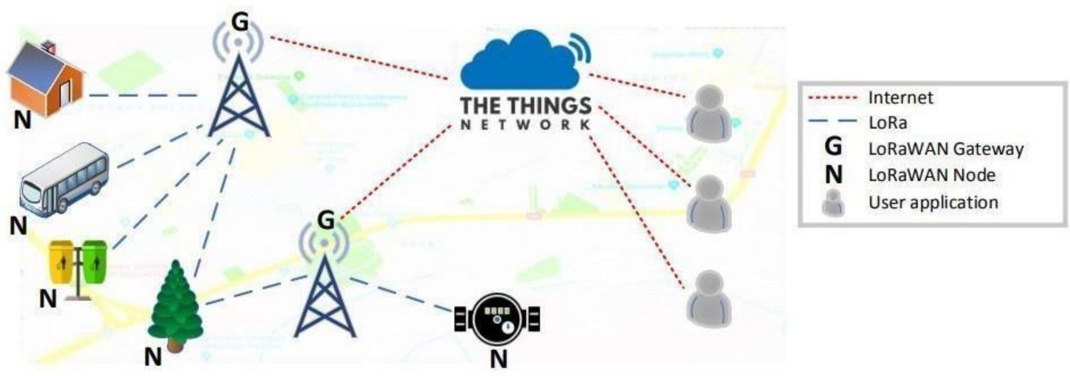

Fig. 2.2: LoRaWAN Gateway

2.6 Working of LoRaWAN Gateway

LoRaWAN gateway can serve multiple groups of devices at a time, although gateways are often deployed in overlapping groups. Devices will send out their signals as RF packets to be picked up by any gateway in range, with the strongest device to gateway connection passing the message on to the cloud. Having multiple gateways adds resilience to a network should one of them fail.

LoRaWAN gateways can be installed and run for a private network controlled by a company. For example, you may manage communications at a reservoir and need to measure water levels or water pressure at specific locations on the site. Physically collecting data from remotely located water level sensors could be difficult, time consuming and costly, but if your sensors are wirelessly connected with LoRaWAN then data can be collected easily and delivered directly to laptop and mobile apps. To do this the sensor will need to wirelessly pass the data first to a LoRaWAN gateway. The gateway will then pass the data from the multiple sensor array on to the cloud and ultimately to your app.

In some, mostly urban locations, LoRaWAN network providers may offer gateway connectivity as a service to multiple end users for a fee. Once picked up by a gateway, RF signals are converted to a format that allows for faster transfer rates. The maximum data rate of LoRa, 50Kb/s, is enough for device to device or device to gateway communication but would prove a serious bottleneck when trying to deliver thousands of messages to the cloud - hence the need for a well-designed gateway to enable higher levels of data transfer from gateway to cloud.

Usually, a connection between a gateway and the cloud is established with ethernet for speed but LTE and Wi-Fi are also practical substitutes in outdoor locations. The data then becomes available in the cloud on whichever application is required. Many sensors will come with their own app to display data in a useful way to the end user. For example, Delta-T's soil moisture sensors use their Delta Link app to show data in graphs and tables making it easy to interpret information and make informed decisions. To better contextualize this, LoRaWAN networks follow a mailbox system.

Gateways act as a mailbox between devices and the cloud, holding, but not reading, messages that go between them. The timing of messages depends on how often an end-device wakes up to send and receive data. If the cloud sends a message to a device, a gateway will hold the message until the device next wakes up.

Gateways are routers equipped with a LoRa concentrator, allowing them to receive LoRa packets. There are two kinds of gateways:

- Low-cost and easy to use gateways that run on a minimal firmware, (e.g., The Things Gateway), run only the packet forwarding software.
- Gateways that run an operating system, for which the packet forwarding software is run as a background program (e.g., Wyld Gateway, Kerlink IoT Station, MultiTech Conduit). These provide more flexibility to the gateway administrator allowing management of the gateway control over install of their own software.

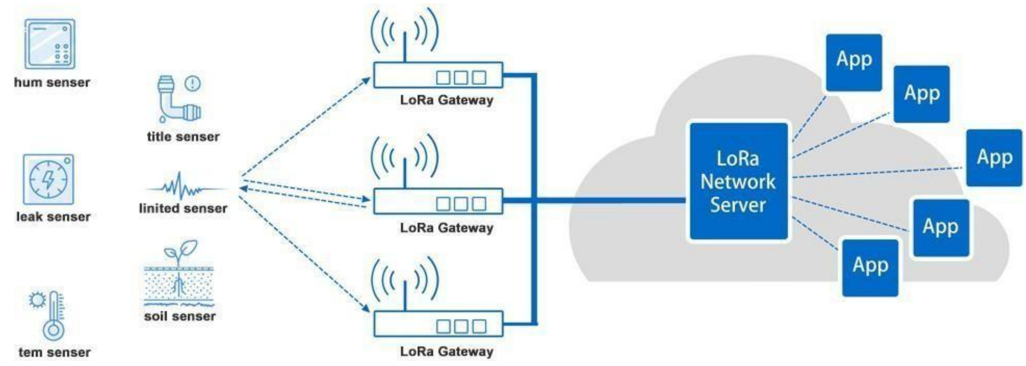

Fig. 2.3 LoRaWAN Gateway in typical LoRaWAN Network.

2.7 Types of LoRaWAN Networks:

There are 2 types of LoRaWAN Networks, they are:

- Public LoRaWAN Network
- Private LoRaWAN Network

Public LoRaWAN Networks: LoRaWAN networks are publicly available in 137 countries under 157 different operators. Joining one of these operators is how many IoT deployments start, although as explained below, setting up your own network is possible, often preferable and sometimes necessary.

- The coverage map above from Lora Alliance shows the countries where an open community network is available. The exact level of coverage varies and is far from 100% in any country. Geography is a key factor, and currently more rural and remote areas are unlikely to be covered even in developed nations.
- Using a public network comes with many advantages. Firstly, the coverage is often quite wide, especially in urban environments, so setting up in range of gateway is often achievable.
- The second is how quick and accessible a public network is. Less infrastructure is required as a user so you can focus on what you do best - the end-devices that collect the data.
- LoRa is also inexpensive as it runs on an unlicensed band, meaning licenses don’t have to be paid to a government authority by a network provider as they are with 4G and 5G. With an established provider, costs of using the network are therefore kept low, making it an affordable solution.

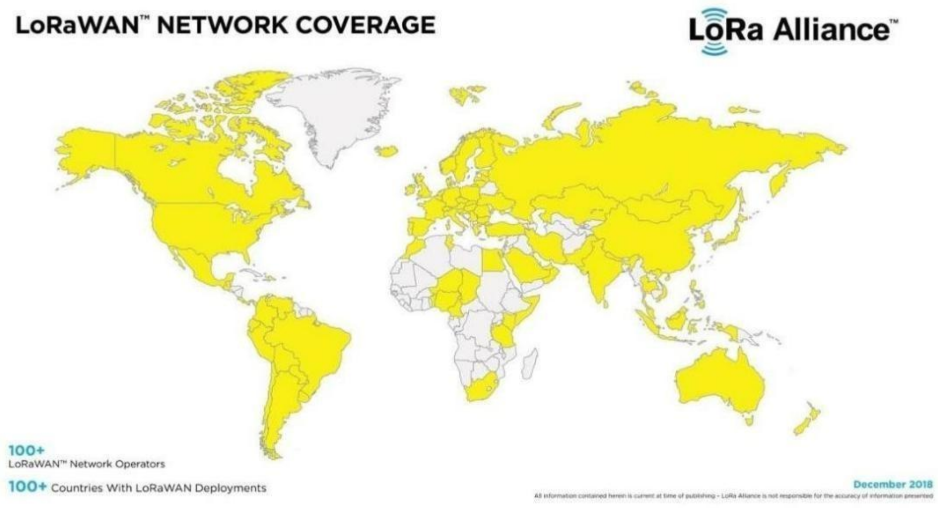

Fig. 2.4 LoRaWAN Network coverage over the world.

Private LoRaWAN Networks: Running your own gateway is a little more complicated. You need to know the details of how to configure the network and what software it should run, but in return it provides a private network limited to your organization or others you share it with.

- This has upsides in that you can easily monitor traffic and have the usual benefits associated with not relying on a third party. You have control over the location of the gateway which can increase efficiency and mean you can work even in the most remote or difficult to reach locations.
- Running a private LoRaWAN network also gives control on capacity per gateway and comes with the option of loading your own firmware to make it truly specialized at its task. A private LoRaWAN gateway generally allows more edge processing flexibility and capability
- There are also implications for security. You may not want your data being controlled by an external provider. Running a localized closed system is ideal for this. Private LoRaWAN gateways allow increased control and security.
- LoRaWAN Gateways are the link between the source of data - sensors in the environment - and people. They enable scale for the Internet of Things and are a key element in the chain that enables better visibility, reduction in costs, reduction in resource inputs, improved safety and better decision making.

2.8 Various LoRaWAN Gateways:

There are various types of LoRaWAN gateways for various purposes where some of them are:

1. **WisGate LoRaWAN Gateway:**
   Features:
   - MT7628, DDR2 RAM 128MB computing.
   - Wi-Fi feature: 2.4GHz (802.11b/g/n) frequency, -95dBm (min) RX sensitivity, 20dBm (max) TX power.
   - LoRa feature: SX1302 Mini PCIe card, 8 channels, -139dBm (min) RX sensitivity.
   - Cellular: Supports Quectel EG95-E/EG95-NA (IoT/M2M -optimized LTE Cat 4 Module).

   Price: INR 45,456/- + GST

2. **Arduino Pro Gateway Edge 2:**
   Features:
   - High transmission power
   - Two fiberglass antennas with 5dBi gain
   - Secure Ethernet, Wi-Fi, or LTE connectivity
   - RAK wireless technologies
   - On-pole and DIN-rail installation kit

   Price: INR 27,332/- + GST

3. **Dragino Multi Channel LPS8:**
   Features:
   - One RJ45 Ethernet port that supports 10/100 Mbps
   - One 2.4 GHz 802.11bgn radio
   - One USB host port
   - Power supply: USB Type-C: 5V, 2A

   Price: INR 26,999/- + GST

4. **Seed Studio 114992857 Gateway:**
   Features:
   - High-performance Cortex A8 1GHz processor
   - Linux system
   - Support for multiple ISM bands
   - Support for multiple methods to access the network, including 4G, Ethernet, and Wi-Fi
   - Support for professional management tools and cloud services

   Price: INR 31,613/- + GST

Fig. 2.5: Various LoRaWAN Gateways.

2.9 Arduino as a Low-Cost LoRaWAN Gateway

The Arduino Uno is a widely used open-source microcontroller board designed for building embedded systems and interactive electronic projects. It is based on the ATmega328P, which acts as the brain of the system by processing inputs and controlling outputs. The microcontroller executes instructions stored in its internal memory. It operates using a clock signal (16 MHz) to synchronize operations. Input signals are processed through ADC (Analog-to-Digital Conversion). Output signals can be digital (HIGH/LOW) or PWM for analog-like control. Communication protocols like UART, SPI, and I2C enable device interfacing. It serves as a platform for rapid prototyping and embedded system design.

Fig. 2.6: Arduino UNO.

A Arduino UNO can be an excellent and cost-effective option for building a LoRaWAN gateway, allowing you to create a customizable and versatile gateway solution for your IoT projects. By leveraging Raspberry Pi's computing power and connectivity options, you can set up a LoRaWAN gateway that enables communication between your IoT devices and the LoRaWAN network server.

Features of Arduino UNO as LoRaWAN Gateway:

When using a Arduino UNO as a LoRaWAN gateway, you can leverage its various features and capabilities to create an efficient and customizable gateway solution for your IoT projects. Some of the key features of using a Raspberry Pi as a LoRaWAN gateway include:

- **Cost-Effectiveness:** Raspberry Pi boards are relatively affordable, making them a budget-friendly option for building LoRaWAN gateways, especially for small-scale or DIY projects.
- **Customizability:** Raspberry Pi's open-source nature allows for extensive customization and configuration, enabling you to tailor the gateway setup to your specific requirements and preferences.
- **Processing Power:** Raspberry Pi boards are equipped with sufficient processing power to handle the data processing and communication tasks required for LoRaWAN gateways, ensuring smooth data transmission and reception.
- **Connectivity Options:** Raspberry Pi boards come with various connectivity options, including Ethernet, Wi-Fi, and Bluetooth, allowing you to connect the gateway to the internet and other devices seamlessly.
- **Expansion Capabilities:** Raspberry Pi boards support the use of various add-on modules and peripherals, providing the flexibility to expand the gateway's capabilities and integrate additional features as needed for your IoT applications.
- **Community Support:** The Raspberry Pi community is extensive and active, offering access to a wealth of resources, tutorials, and forums where you can find guidance and assistance for setting up and troubleshooting your LoRaWAN gateway project.
- **Compatibility with LoRaWAN Concentrator Boards:** Raspberry Pi boards can be easily paired with compatible LoRa concentrator boards, such as those based on the Semtech SX1301 or SX1308 chipset, enabling seamless integration with LoRaWAN networks.
- **Software Compatibility:** Raspberry Pi supports a wide range of software applications and operating systems, allowing you to choose from various LoRaWAN gateway software options that best suit your project requirements.
- **Low Power Consumption:** Raspberry Pi boards are designed to be energy-efficient, consuming relatively low power while providing sufficient computing power for running the LoRaWAN gateway and other related applications.
- **Versatility:** Raspberry Pi boards can be used for various IoT and computing projects beyond just serving as LoRaWAN gateways, making them a versatile and multipurpose platform for experimentation and development.

CHAPTER – III
E-BYTE E32-433T30D LoRa MODULE

CHAPTER – 3
E-BYTE E32-433T30D LoRa MODULE

3.1 Introduction to E-Byte E32-433T30D LoRa Module

E32-433T30D is adopts LoRa spread spectrum technology, with this technology the module can transmit much longer distance than similar products under the same power with better anti-interference performance. The E32-433T30D LoRa module is a popular wireless communication module widely used for long-range, low-power communication in various applications such as the Internet of Things (IoT), smart agriculture, smart cities, and industrial automation. Developed by Ebyte, a leading provider of IoT solutions, the E32-433T30D module operates in the 433MHz frequency band, offering an impressive communication range of up to 3000 meters in open space, making it suitable for long-range applications.

Key features of the E32-433T30D LoRa module include its high sensitivity, low power consumption, and robust performance in challenging environments. With its LoRa modulation technology, the module provides reliable communication even in areas with obstacles or interference, making it an ideal choice for applications that require a stable and long-distance wireless connection. Furthermore, the E32-433T30D module is designed with a simple interface, making it easy to integrate into various electronic devices and systems. Its UART interface enables seamless communication with microcontrollers, allowing for convenient data transmission and reception. Additionally, the module supports a wide range of configurations, allowing users to customize parameters such as the communication frequency, transmission power, and air data rate to suit specific application requirements.

One of the notable advantages of the E32-433T30D module is its low power consumption, which is vital for battery-operated devices and applications that prioritize energy efficiency. This feature makes it well-suited for IoT devices that need to operate for extended periods without frequent battery replacements or recharging. Moreover, the E32-433T30D LoRa module complies with various industry standards, ensuring its compatibility with different LoRaWAN networks and protocols. This compliance facilitates the seamless integration of the module into existing IoT infrastructures and networks, making it easier for developers and engineers to incorporate the module into their projects without significant modifications.

Fig. 3.1: E-Byte E32-433T30D LoRa Module.

3.2 Features

1. Communication distance tested is up to 8km
2. Maximum transmission power of 1W, software multi-level adjustable
3. Support the global license-free ISM 433MHz band
4. Support air date rate of 0.3kbps～19.2kbps
5. Support new generation LoRa technology.
6. Low power consumption for battery supplied applications；
7. Support 3.3V~5.2V power supply, power supply over 5.0 V can guarantee the best performance
8. Industrial grade standard design, support -40 ~ 85 °C for working over a long time
9. SMA access point, good for secondary development and integration.

3.3 Module Layout and Pin Description.

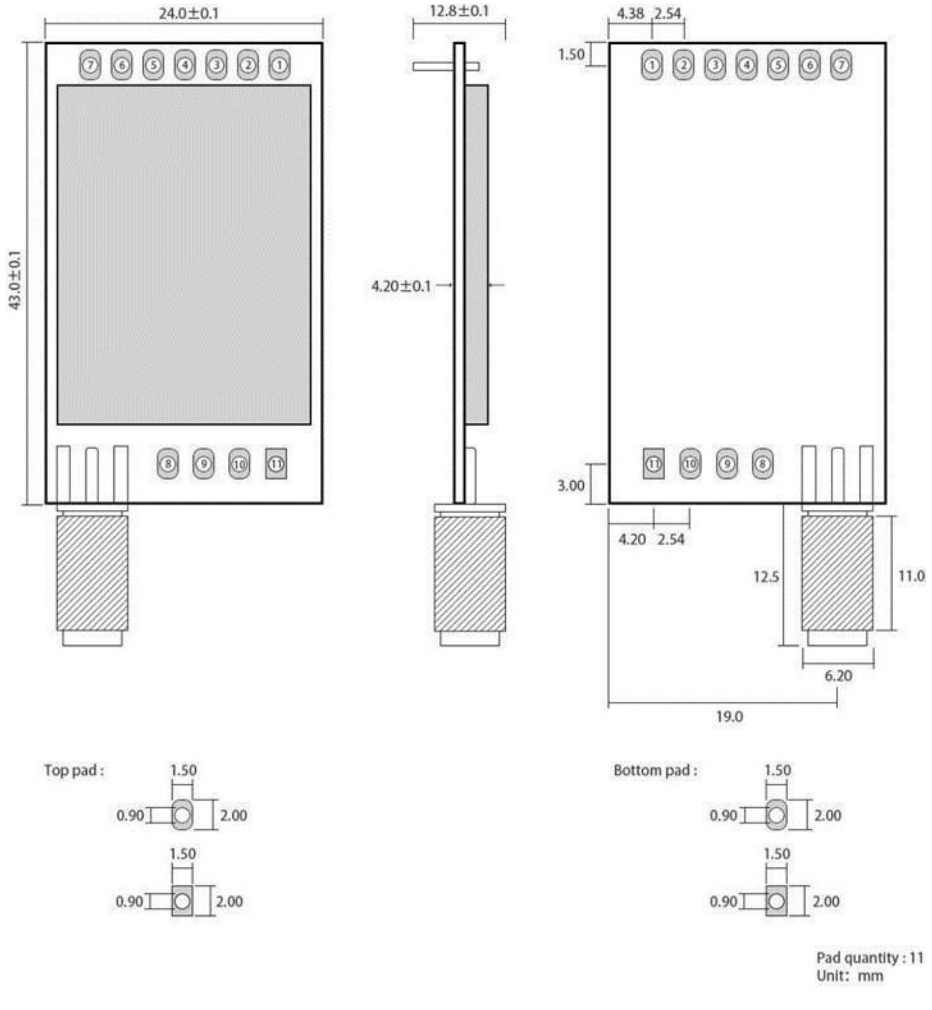

Fig. 3.2: Size and Layout of E-Byte E32-433T30D LoRa Module.

No.	Name	Direction	Function	
1	M0	Input (weak pull-up)	Work with M1 to decide 4 working modes of module (not suspended, if not used, could be grounded).	
2	M1	Input (weak pull-up)	Work with M0 to decide 4 working modes of module (not suspended, if not used, could be grounded).	
3	RXD	Input	TTL UART inputs, connects to external (MCU, PC) TXD output pin. Can be configured as open-drain or pull-up input.	
	TXD	Output	TTL UART outputs, connects to external RXD (MCU, PC) input pin. Can be configured as open-drain or push-pull output	
5	AUX	Output	To indicate module' s working status & wakes up the external MCU. During the procedure of self-check initialization, the pin outputs low level. Can be configured as push-pull output	
6	VCC	Input	Power supply ： 3.3~ 5.2V DC	
7	GND	Input	Ground	

Table. 3.1: Pin Description of E-Byte E32-433T30D LoRa Module.

Fig. 3.3: Pin Layout of E-Byte E32-433T30D LoRa Module.

3.4 Operating Parameters

Main parameter	Performance			Remark
	Min	Type	Max	
Operating voltage (V)	3.3	5.0	5.2	≥5.0 V ensures output power
Communication level (V)	-	3.3	-	For 5V TTL, it may be at risk of burning down
Operating temperature (°C)	-40	-	85	Industrial design
Operating frequency (MHz)	410	-	441	Support ISM band
Transmitting current [mA]	-	610	Instant Power Consumption	Instant power consumption
Receiving current [mA]	-	20	-	-
Turn-off current [μA]	-	5	-	Software is shut down
Max Tx power (dBm)	29.5	30	30.5	-
Receiving sensitivity (dBm)	-145	-147	-148	Air data rate is 2.4kbps
Air data rate (bps)	0.3k	2.4k	19.2k	Controlled via user's programming

Table. 3.2: Operating Parameters of LoRa E-Byte E32-433T30D LoRa Module.

Main parameter	Description	Remark
Distance for reference	8000m	Test condition： clear and open area, antenna gain: 5dBi, antenna height: 2.5m， air data rate: 2.4kbps
TX length	58 Byte	Maximum capacity of single package
Buffer	512 Byte	-
Modulation	LoRa™	-
Communication interface	TTL	@3.3V
Package	DIP	-
Connector	2.54mm	-
Size	24 * 43mm	-
Antenna	SMA-K	50ohm Impedance

Table. 3.3: Main parameters of LoRa E-Byte E32-433T30D LoRa Module.

3.5 Spread Spectrum Communication

LoRa (Long Range) spread spectrum modulation is a robust and efficient wireless communication technology known for its exceptional long-range capabilities and its ability to operate in environments with high levels of interference. LoRa technology employs a type of chirp spread spectrum modulation, which allows for a significant increase in the communication range while maintaining low power consumption. This method of spread spectrum modulation enables LoRa devices to achieve an extended communication range of several kilometres, making it well-suited for various applications that require long-range and low-power wireless connectivity.

One of the key advantages of LoRa spread spectrum communication is its remarkable resistance to interference from other wireless devices operating in the same frequency band. The spread spectrum technique enables LoRa signals to spread over a wide frequency band, making them less susceptible to narrowband interference. This characteristic enables LoRa devices to maintain reliable communication even in environments with high levels of electromagnetic noise, such as urban areas or industrial settings, where other wireless technologies may struggle to provide stable connectivity.

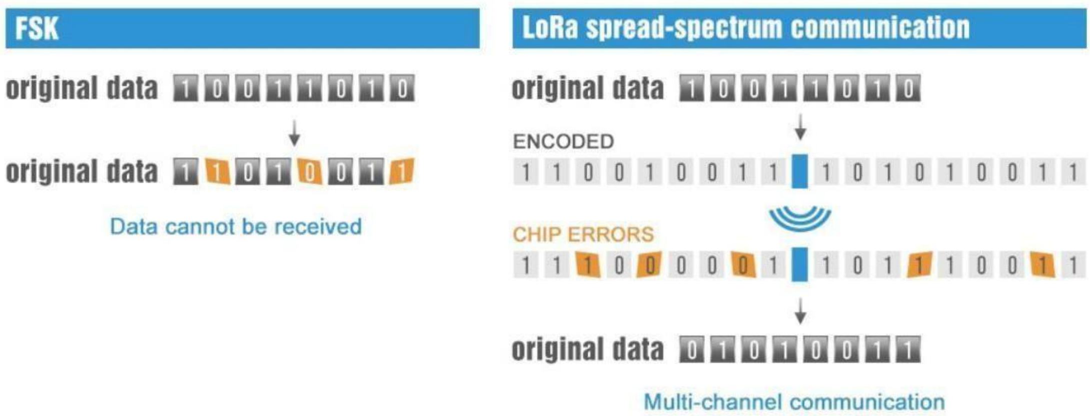

Fig. 3.4: LoRa Spread Spectrum Communication.

Forward Error Correction:

Forward Error Correction (FEC) is a method used in data transmission to detect and correct errors that occur during the transmission process. It is a technique that adds redundancy to the transmitted data, enabling the receiver to identify and correct errors without the need for retransmission. FEC plays a crucial role in ensuring reliable communication over unreliable or noisy channels, making it an essential component in various communication systems and protocols.

The fundamental principle behind FEC is to introduce extra redundant bits into the data stream before transmission. These redundant bits contain information that allows the receiver to detect and rectify errors that may occur during transmission. By encoding the data with additional redundant information, FEC enables the receiver to reconstruct the original data accurately, even if some bits are corrupted or lost during transmission.

There are different types of FEC techniques, each with its own encoding and decoding algorithms. Some common FEC methods include:

- **Block codes:** These divide data into fixed-size blocks and add parity bits to each block for error detection and correction.
- **Convolutional codes:** These are based on shift registers and generate redundant bits based on the previous input bits.
- **Reed-Solomon codes:** These are particularly effective for correcting burst errors and are widely used in data storage and transmission systems.

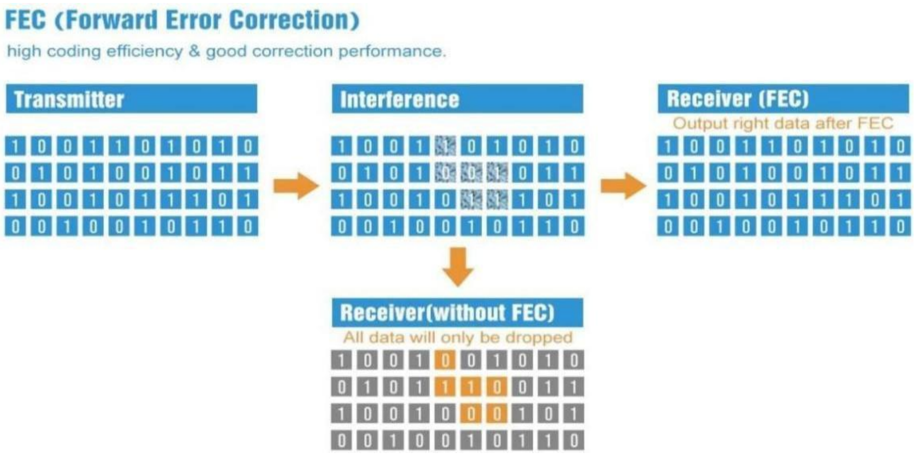

Fig. 3.5: Forward Error Correction.

3.6 Interfacing Module with Microcontroller Unit

Interfacing the Ebyte E32-433T30D LoRa module with a microcontroller unit (MCU) is a crucial step in incorporating the module into various IoT and communication projects. To effectively establish communication between the LoRa module and the MCU, you can follow these general steps:

- **Hardware Connections:**
  Ensure that the necessary hardware connections between the LoRa module and the MCU are properly established. Typically, this involves connecting the appropriate pins of the LoRa module (such as TX, RX, and power supply) to the corresponding pins of the MCU.
- **Voltage Level Compatibility:**
  Verify that the voltage levels of the LoRa module and the MCU are compatible. If there are discrepancies, use level shifters or voltage dividers to ensure that the signals are properly interfaced.
- **Software Configuration:**
  Implement the required software configuration for the MCU to communicate with the LoRa module. This usually involves setting up the UART communication protocol,

which is commonly used for serial communication between the MCU and the LoRa module.

- **UART Communication:**
  Configure the UART communication parameters, such as baud rate, data bits, stop bits, and parity bits, to match the settings of the E32-433T30D LoRa module. Ensure that the communication parameters are consistent between the MCU and the LoRa module to enable seamless data transfer.
- **Data Transmission and Reception:**
  Implement the necessary code in the MCU to facilitate data transmission and reception with the LoRa module. This typically involves sending and receiving commands and data packets through the UART interface, as per the specifications outlined in the datasheet or user manual of the E32-433T30D module.
- **Error Handling and Data Processing:**
  Implement error handling mechanisms and data processing algorithms within the MCU to manage and interpret the data received from the LoRa module. This may involve error-checking protocols and data parsing methods to ensure data integrity and reliability.
- **Testing and Debugging:**
  Conduct thorough testing and debugging procedures to verify the successful integration and communication between the MCU and the Ebyte E32-433T30D LoRa module. Monitor the data exchange, check for any errors or inconsistencies, and make necessary adjustments to the code and hardware connections as needed.

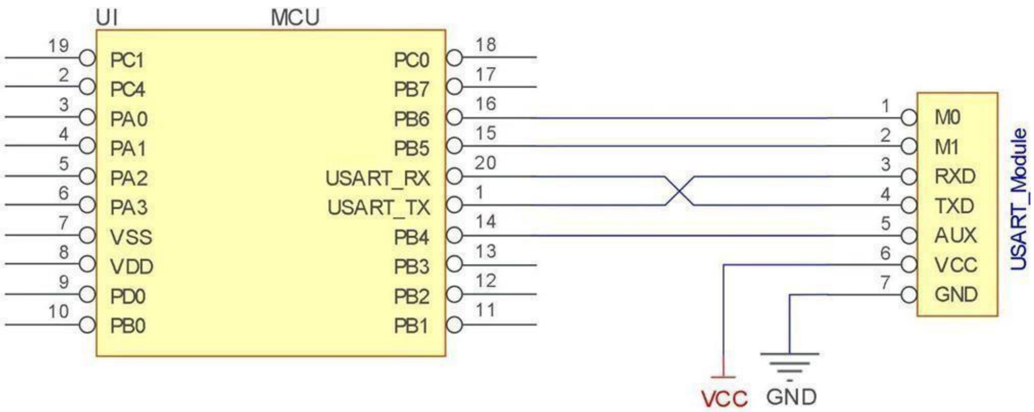

Fig. 3.6: Interfacing the LoRa Module with a Microcontroller using UART.

3.7 Operating Modes

There are four operating modes, which are set by M1 and M0, the details are as follows:

Mode (0-3)	M0	M1	Mode introduction	Remark
0 Normal	0	0	UART and wireless channel are open, transparent transmission is on	The receiver work in must mode 0 or mode 1
1 Wake up	1	0	UART and wireless channel are open, the only difference with mode 0 is that before transmitting data, increasing the wake-up code automatically, so that it can awake the receiver under mode 3.	The receiver could be 0,1 or 2
2 Power	0	1	UART close, wireless is under air-awaken mode, after receiving data, UART open and send data.	Transmitter must be mode 1, unable to transmit in this mode.
3 Sleep	1	1	sleep mode, parameter setting command is receiving available.	more details parameter specification.

Table. 3.4: Operating Modes in LoRa E-Byte E32-433T30D LoRa Module.

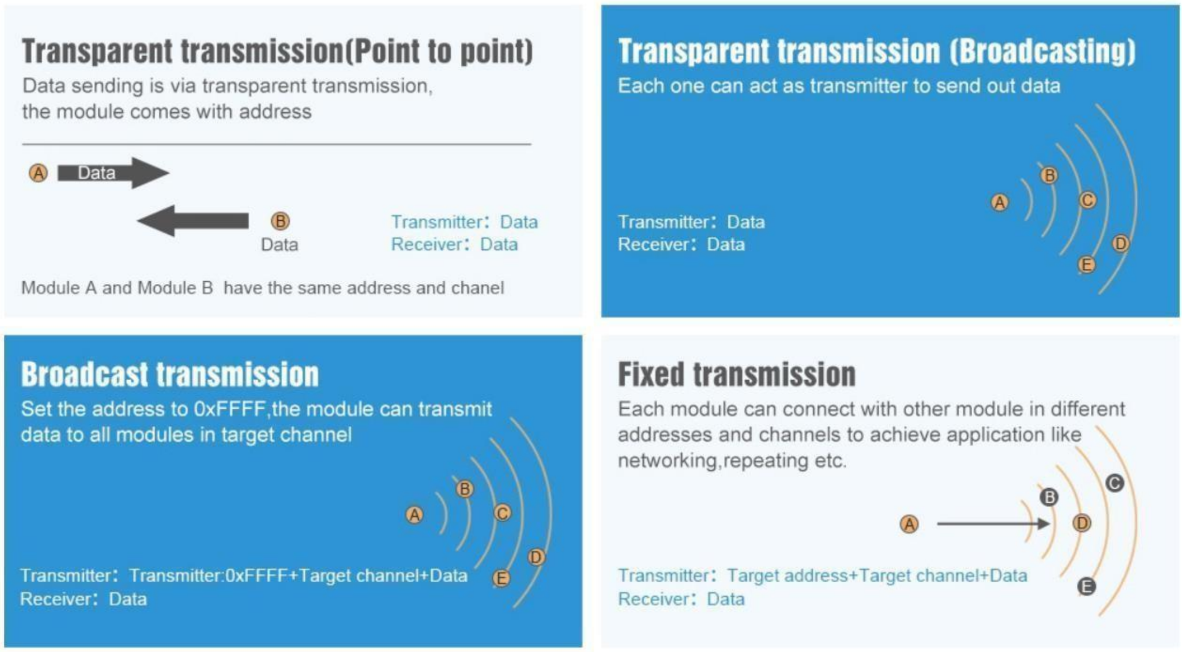

Fig. 3.7: Modes of Transmission in LoRa E-Byte E32-433T30D LoRa Module.

3.8 Hardware Design

- It is recommended to use a DC stabilized power supply. The power supply ripple factor is as small as possible, and the module needs to be reliably grounded.
- Please pay attention to the correct connection of the positive and negative poles of the power supply. Reverse connection may cause permanent damage to the module.
- Please check the power supply to ensure it is within the recommended voltage otherwise when it exceeds the maximum value the module will be permanently damaged.
- Please check the stability of the power supply, the voltage cannot be fluctuated frequently.
- When designing the power supply circuit for the module, it is often recommended to reserve more than 30% of the margin, so the whole machine is beneficial for long-term stable operation.
- The module should be as far away as possible from the power supply, transformers, high-frequency wiring and other parts with large electromagnetic interference.
- High-frequency digital routing, high-frequency analog routing, and power routing must be avoided under the module. If it is necessary to pass through the module, assume that the module is soldered to the Top Layer, and the copper is spread on the Top Layer of the module contact part (well grounded), it must be close to the digital part of the module and routed in the Bottom Layer.
- Assuming the module is soldered or placed over the Top Layer, it is wrong to randomly route over the Bottom Layer or other layers, which will affect the module's spurs and receiving sensitivity to varying degrees.
- It is assumed that there are devices with large electromagnetic interference around the module that will greatly affect the performance. It is recommended to keep them away from the module according to the strength of the interference. If necessary, appropriate isolation and shielding can be done.
- Assume that there are traces with large electromagnetic interference (high-frequency digital, high-frequency analog, power traces) around the module that will greatly affect the performance of the module. It is recommended to stay away from the module according to the strength of the interference. If necessary, appropriate isolation and shielding can be done.
- If the communication line uses a 5V level, a 1k-5.1k resistor must be connected in series (not recommended, there is still a risk of damage).
- Try to stay away from some physical layers such as TTL protocol at 2.4GHz, for example: USB3.0.
- The mounting structure of antenna has a great influence on the performance of the module. It is necessary to ensure that the antenna is exposed, preferably vertically upward. When the module is mounted inside the case, use a good antenna extension cable to extend the antenna to the outside.
- The antenna must not be installed inside the metal case, which will cause the transmission distance to be greatly weakened.

CHAPTER – IV
NODE TO NODE COMMUNICATION

CHAPTER – 4
NODE TO NODE COMMUNICATION

4.1 E-Byte E32-433T30D LoRa Module Testing

The E-Byte E32-433T30D LoRa Module is tested by creating 2 nodes and each node is equipped with a LoRa Module and a microcontroller such as Arduino UNO or NodeMCU or STM32 or a Raspberry Pi Pico. Here the two nodes were classified as both sender and receiver termed as trans receiver. The purpose node to node communication is to find whether the LoRa modules were actually communicating or whether we are experiencing any garbage values while the transmission and reception process.

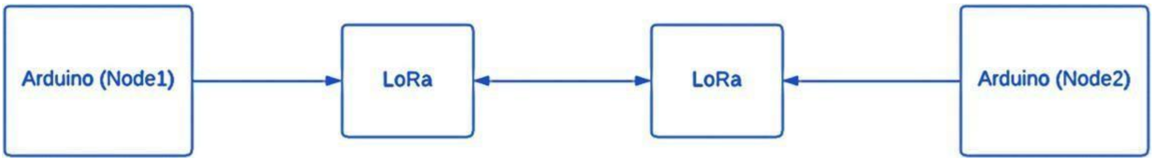

Fig. 4.1 Node to Node Communication using LoRa.

Here the process of testing is we send some text data at the sender end which is one of the 2 nodes where by the help of microcontroller we transmit the data over LoRa and at the receiver node the LoRa will receive the incoming data and display it over Serial Monitor in the receiver end. Here both the LoRa's have same address and channel and they communicate with each other in transparent transmission mode.

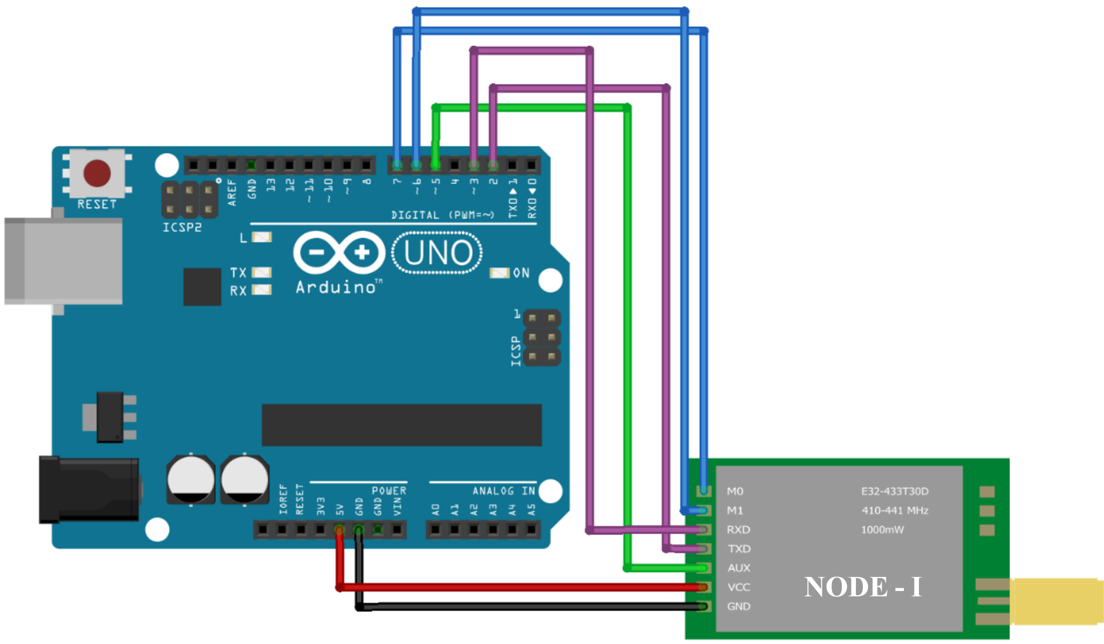

Fig. 4.2: Circuit diagram of LoRa Interfacing with Arduino UNO.

The pin configuration for Arduino UNO and E-Byte E32-433T30D LoRa Module is;

Arduino UNO	E-Byte E32-433T30D LoRa Module
5V	Vcc
GND	GND
D5	AUX
D2	TX
D3	RX
D7	M0
D6	M1

Table. 4.1: Pin Configuration for Arduino UNO and E-Byte E32-433T30D LoRa Module.

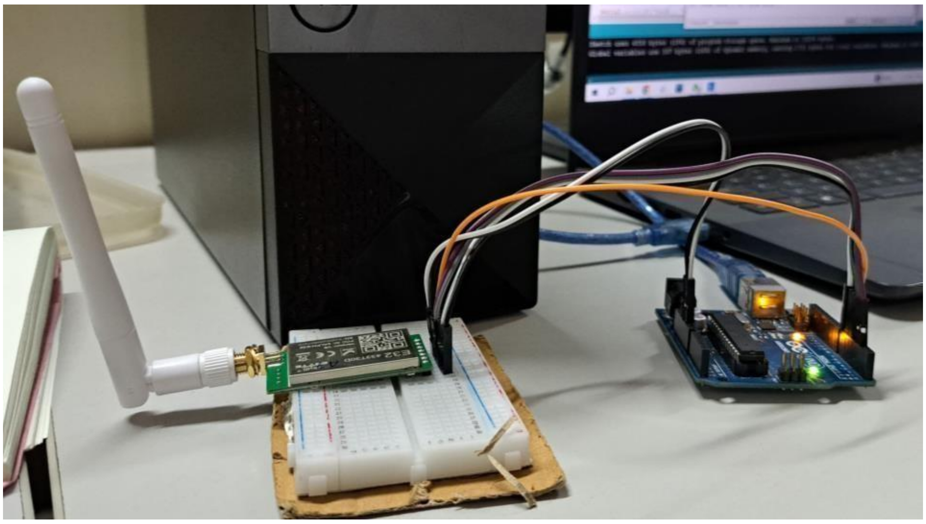

Fig. 4.3: Single LoRa Node.

The results of the test will be obtained when we send some text data over the LoRa communication are;

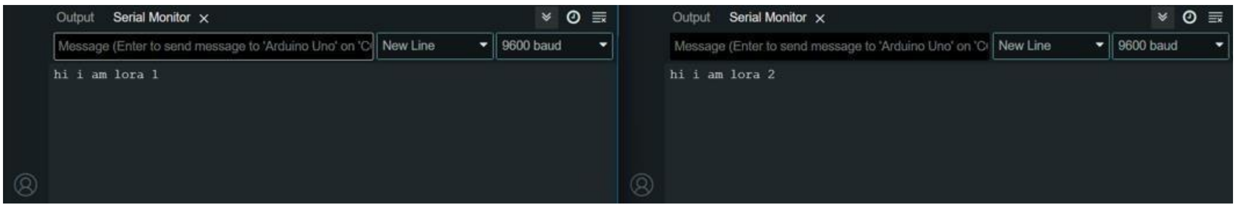

Fig. 4.4: Output of Node-to-Node Communication by LoRa.

4.2 Configuring LoRa Modules

These LoRa Modules are UART interfaced modules so to configure the modules we need a converter called UART to USB converter. Configuring the module is to set the module properties and assign the modules with addresses and channels to the LoRa Modules. Generally, here we can use two modes programming the module by keeping it in operating mode 3 which is configuring mode where we access it through giving M0 and M1 pins of the module High or "1" or we can use a RF – Setting software for accessing the module parameters.

Fig. 4.5 UART to USB Converter.

The interfacing of UART to USB Converter with E-Byte E32-433T30D LoRa Module is;

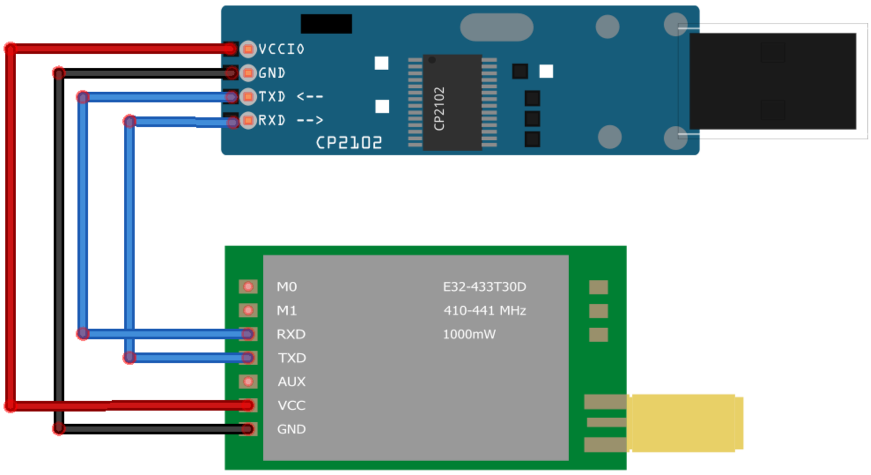

Fig. 4.6 UART to USB Converter Interfacing with E-Byte E32-433T30D LoRa Module.

The pin configuration for UART to USB Converter and E-Byte E32-433T30D LoRa Module is;

UART to USB Converter	E-Byte E32-433T30D LoRa Module
Vcc	Vcc
GND	GND
RX	TX
TX	RX

Table. 4.2: Pin Configuration for UART to USB Converter with E-Byte E32-433T30D LoRa Module.

The RF Setting software for E-Byte E32-433T30D LoRa Module is interactive software where we interface the module with UART to USB Converter and we open the active UART COM port over the available COM ports and we can open the port for accessing the module parameters. These are the following parameters that we will obtain through the RF Setting software.

- UART Rate
- Parity
- Air Data Rate
- Power
- FEC Status
- Fixed Mode Status
- WOR Timing
- I/O Mode Status
- Address
- Channel

Fig. 4.7: RF Setting software Interface.

The parameters that were set to E-Byte E32-433T30D LoRa Module in order to achieve node-to-node communication are;

Parameter	Description
UART Rate	9600 bps
Parity	8N1
Air Data Rate	2.4Kbps
Power	30dBm
FEC	Enable
Fixed Mode	Enable
WOR Timing	250ms
I/O Mode	Push Pull
Address	Range (0-65,535)
Channel	Range (0-23)

Table. 4.3: The parameters configured for E-Byte E32-433T30D LoRa Module.

Assigning Addresses and Channels for the Module:

The E-Byte E32-433T30D LoRa Module has vast range of addresses as (0-65,535) and channels from (0-23). Multiple Nodes can have same address or channel number where the mode of transmission changes node to node. As the wide range of addresses indicates the scope of scalability in the network.

The channel numbers in the LoRa decides the frequency at which the module should transmit the data. The E-Byte E32-433T30D LoRa Module works in a bandwidth of 410 – 433 MHz where it is divided into 23 channels and we can assign the channel numbers in the range of (0-23) where the frequency of a particular channel is the some of lower frequency which is 410 MHz and the channel number. If we assign channel number as 10 then the data will transmit at frequency 420MHz.

4.3 UART Interface

UART means Universal Asynchronous Receiver Transmitter Protocol. UART is used for serial communication from the name itself we can understand the functions of UART, where U stands for Universal which means this protocol can be applied to any transmitter and receiver, and A is for Asynchronous which means one cannot use clock signal for communication of data and R and T refers to Receiver and Transmitter hence UART refers to a protocol in which serial data communication will happen without clock signal.

UART is established for serial communication. In this article, we will discuss how parallel communication is established with respect to serial communication using UART as well as how to configure UART and what is the data format in UART. Later on, we will discuss the Pros and Cons of the UART.

UART is very simple and only uses two wires between transmitter and receiver to transmit and receive in both directions. Both ends also have a ground connection. Communication in UART can be simplex (data is sent in one direction only), half-duplex (each side speaks but only one at a time), or full-duplex (both sides can transmit simultaneously). Data in UART is transmitted in the form of frames. The format and content of these frames is briefly described and explained.

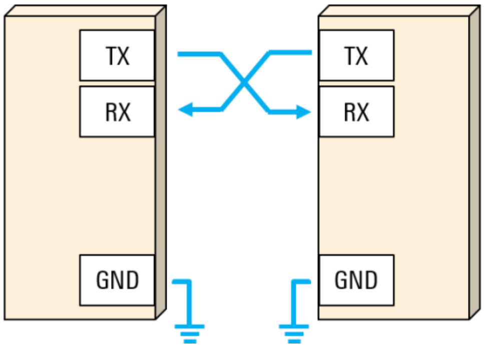

Fig. 4.8: UART Interfacing.

UART is a Universal Asynchronous Receiver Transmitter protocol that is used for serial communication. Two wires are established here in which only one wire is used for transmission whereas the second wire is used for reception. Data format and transmission speeds can be configured here. So, before starting with the communication define the data format and transmission speed. Data format and transmission speed for communication will be defined here and we do not have a clock over here that's why it is referred to as asynchronous communication with UART protocol. Here we will see how this protocol is designed physically.

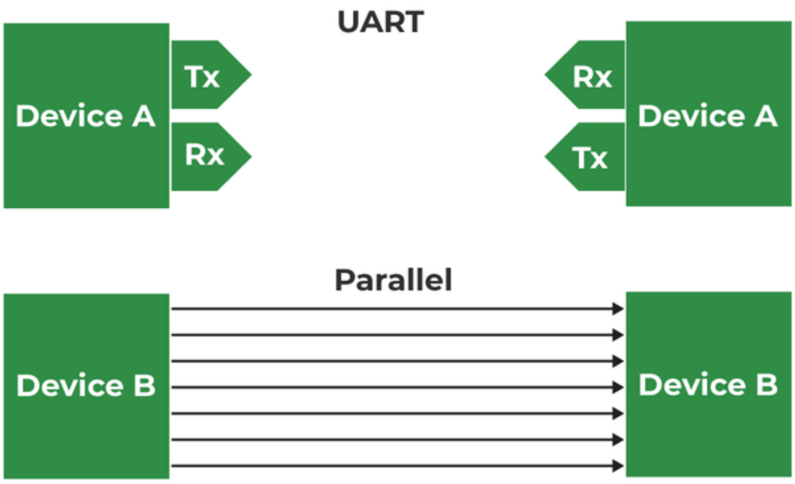

Fig. 4.9: UART Interfacing between two devices.

Here, DEVICE A that is having transmitter PIN and a receiver pin; DEVICE B is having a receiver and transmission pin. The Transmitter of DEVICE A is one that should be connected with the receiver pin of DEVICE B and the transmitter pin of DEVICE B should be connected with the receiver pin of DEVICE A we just need to connect two wires for communication.

If DEVICE A wants to send data, then it will be sending data on the transmitter's pin and here receiver of this DEVICE B will receive it over and if DEVICE A wants to receive the data, then that is possible on the RX line that will be forwarded by TX of DEVICE B. On comparing this serial communication of UART with parallel then it can be observed that in parallel multiple buses are required. Based on the number of lines bus complexity of UART is better but parallel communication is good in terms of speed. So, when speed is required at that time, we should select parallel communication and for a low-speed application, UART must be used and the bus complexity will be less.

The configuration of UART is done before transmission, both of these devices are connected with protocol and should know the speed of data transmission. First, define the speed of both devices. Now, configure the speed of DEVICE A and B for data transmission which is referred to as Baud Rate so here Baud Rate will be the same for DEVICE A and B otherwise both of these devices cannot understand at what speed and at what rate data is coming. After that, configure the data length so here DEVICE A and DEVICE B both are configured at fixed data length if DEVICE A is transmitting data, then it is configured with fixed data. Like if DEVICE A is configured with the eight-bit size of data then DEVICE B should also be configured at the same size of data which is eight bits. After this, check data transmission or receiving time, forward start bits, and stop bits.

Suppose DEVICE A is sending data to DEVICE B, and the transmitter of DEVICE A will send data to the receiver of Device B then it will be logic high. Now, send the start bit that will be logic 0 and once we have the start bit, DEVICE B will know that somebody is communicating. Now there is the same speed configuration with both devices. So, after the start bit, DEVICE A can forward data.

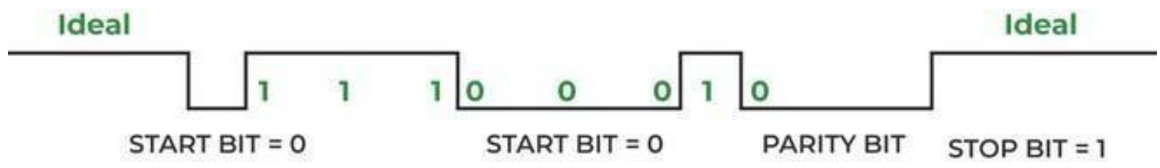

Fig. 4.10: UART data format.

Consider 8 bits of data length, so we will be forwarding 8 bits and those 8 bits will be received by DEVICE B a parity bit can also be used which is optional, but this is quite effective. By using the parity bit, it can be identified whether the received data is correct or not. Suppose we are sending 1 1 1 0 0 0 1 0. Now, we have 4 ones; an even number of ones are there hence the parity is even and for that, logic 0 will be assigned. Suppose we are receiving data with some error, say zero is converted into one; Now incorrect data that is 1 1 1 1 0 0 1 0 for this incorrect data parity will be 0 as there are 5 ones, here is a mismatch in the parity bit and hence it is confirmed that the received data has some error.

Pros of UART Protocol:

- It is having less physical interfacing based on only two lines.
- Simple to configure data and data size. Speed is also configurable. In the majority of cases, this baud rate is 9600 for the UART protocol. Full duplex mode configuration is possible by using two wires so that is the major advantage of UART.
- Error detention is possible

Cons of UART Protocol:

- UART is having serial communication, therefore, it has less speed.
- Start bit, stop bit, and the parity bit is other overhead.
- Since this is asynchronous communication so here there are many things that we need to do in configuration, for instance, we should configure both devices at the same speed because the clock signal is absent.

4.4 Distance Testing of E-Byte E32-433T30D LoRa Module

Distance testing for the E-Byte E32-433T30D LoRa module involves measuring the communication range between two or more modules in real-world conditions. This type of testing helps determine the effective range of the LoRa module under specific environmental and operational constraints. The E-Byte E32-433T30D LoRa Module are designed to transmit data over 8Km in Line of Sight.

The distance test of the module in a closed building where both nodes are on the same floor and different floors are conducted and the results are of in a very closed environment like the closed building one, we can achieve up to 100 mts range. In outside environment the distance test of the module has been conducted where we got a range of 600 mts.

Fig. 4.11: Distance Testing of E-Byte E32-433T30D LoRa Module.

4.5 Two Nodes to Master Node Communication Testing

The two nodes which are equipped with LoRa will acts as sender where they will transmit the data to receiver here the 2 nodes are interfaced with 2 microcontrollers and at the receiver end node is also interfaced with a microcontroller. As the both nodes sends the data to the master node at once.

Here we are using the fixed transmission mode of transmitting the data from sender to the receiver. All the modules will be assigned addresses and given same channel number in order to transmit data over same channel.

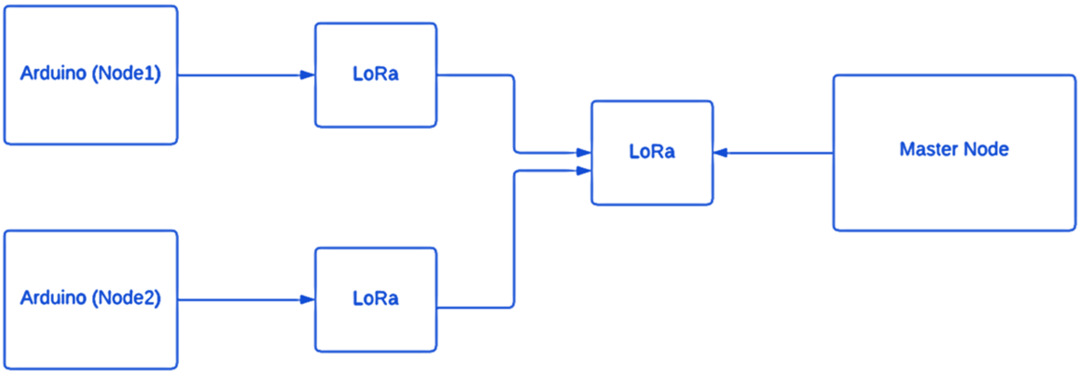

Fig. 4.12: Two Node to Master Node Communication.

Addresses and Channels assigned for nodes:

- **For Sender Node1:**
  - Address = 1
  - Channel = 2
- **For Sender Node2:**
  - Address = 5
  - Channel = 2
- **For Receiver Node:**
  - Address = 7
  - Channel = 2

So, the two LoRa Sender nodes are assigned by their respective addresses and channel numbers and the receiver node LoRa is given with some address and the same channel number as the sender nodes are been assigned in order to transmit data over same channel.

Fig. 4.13: Two LoRa Sender Nodes.

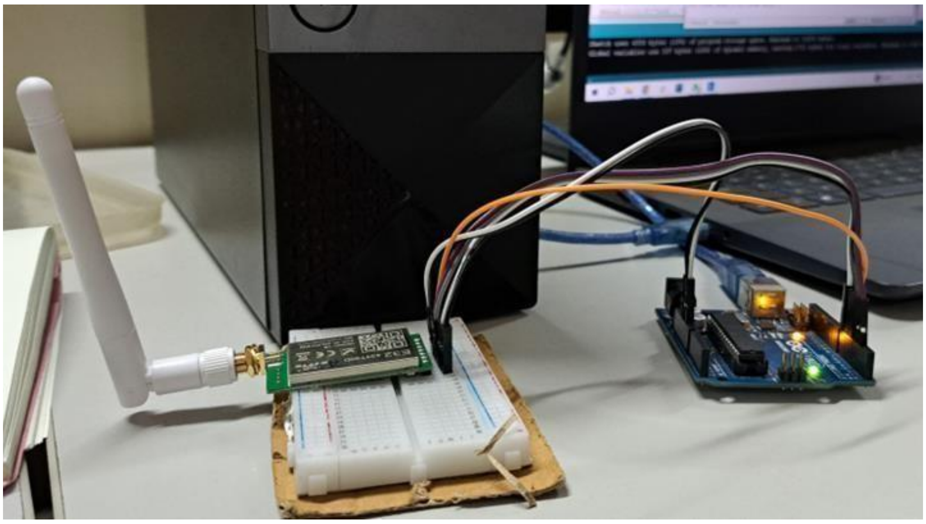

Fig. 4.14: The Receiver LoRa Node.

```
Sender 2
Sender 1
Sender 2
Sender 1
Sender 2
Sender 1
Sender 2
```

Fig.4.15: Output in Serial Monitor.

CHAPTER – V
RSSI Vs DISTANCE AND PATH LOSS Vs DISTANCE ANALYSIS

CHAPTER – 5
RSSI Vs DISTANCE AND PATH LOSS Vs DISTANCE ANALYSIS

5.1 Received Signal Strength Intensity (RSSI)

RSSI stands for Received Signal Strength Indicator. It's a measurement of how well a device can hear a signal from a router or mobile phone. RSSI values indicate the strength and quality of the wireless signal being received. RSSI values are indicated by negative dBm values. The higher the number, the better the signal. The best signal you can receive is -30 dBm, and -90 dBm is unusable. RSSI, or Received Signal Strength Indicator, is a metric used to measure the power level of a received radio frequency (RF) signal. It provides an indication of how strong or weak a received wireless signal is, typically expressed in decibels (dBm) or a relative scale.

- **Measurement Unit:**
  - RSSI is usually measured in decibels (dBm), which is a logarithmic unit representing the power ratio in decibels relative to 1 milliwatt.
- **Signal Strength Interpretation:**
  - A higher RSSI value generally indicates a stronger signal, while a lower value suggests a weaker signal.
  - RSSI values closer to 0 dBm represent a stronger signal, while values further from 0 dBm indicate a weaker signal.
- **Range and Thresholds:**
  - The RSSI value can be used to estimate the distance between a transmitting device (e.g., Wi-Fi router, Bluetooth device) and the receiving device.
  - Different wireless technologies may have different RSSI thresholds for signal quality (e.g., in Wi-Fi, a higher RSSI is generally better).
- **Limitations:**
  - RSSI alone may not provide a complete picture of the link quality or reliability, as it does not consider factors like interference, noise, or the quality of the received signal.
  - Different devices and manufacturers may use different algorithms and calibrations to determine RSSI, leading to variations in reported values.
- **Applications:**
  - Wireless Networks (Wi-Fi): RSSI is commonly used in Wi-Fi networks to assess the signal strength between a device and an access point.
  - Bluetooth: RSSI is utilized in Bluetooth technology to estimate the distance between paired devices.
  - Wireless Sensor Networks: In IoT applications, RSSI can be used to optimize communication between sensor nodes.
  - Localization: RSSI data can be employed for location estimation in some applications.

- **Improvements and Considerations:**
  - Some wireless technologies provide additional metrics beyond RSSI to enhance signal quality assessment (e.g., Wi-Fi may use Signal-to-Noise Ratio, SNR).
  - Combining RSSI with other metrics can offer a more comprehensive evaluation of the wireless environment.
- **Influence of Obstacles and Interference:**
  - RSSI values can be affected by obstacles, reflections, and interference in the signal path, leading to fluctuations in signal strength.
- **RSSI in Wireless Security:**
  - In wireless security, monitoring RSSI can be used to detect unauthorized access points or devices.

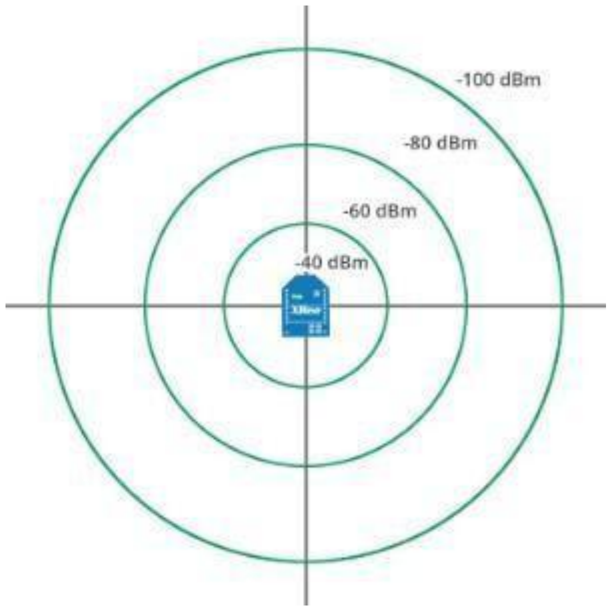

Fig. 5.1 Varying RSSI with Distance.

RSSI	Signal strength	Description
>= -70 dBm	Excellent	Strong signal with maximum data speeds
-70 dBm to -85 dBm	Good	Strong signal with good data speeds
-86 dBm to -100 dBm	Fair	Fair but useful, fast and reliable data speeds may be attained, but marginal data with drop-outs is possible
< -100 dBm	Poor	Performance will drop drastically
-110 dBm	No signal	Disconnection

Fig. 5.2 RSSI Vs Signal Strength.

5.2 Calculation of RSSI with the Distance

The Received Signal Strength Intensity (RSSI) will be calculated by using a simple formula;

- **RSSI (dB) = Transmit Power(dB) + Antenna Gain(dB) – Path Loss(dB)**
- **Transmit Power = 30 dBm ➔ 10*log (30) dB ➔ 14.771dB**
- **Antenna Gain = 3.6dBi**
- **As (dB = dBi – 2.15) ➔ Antenna gain = 3.6 – 2.15 = 1.45dB**

After calculating the RSSI for the distances of range (0.1Km to 4Km) we obtain the values of RSSI in dB as;

RSSI Vs Distance								
S.No	Distance (Kms)	Frequency (MHz)	Transmit Power(dBm)	Transmit Power(dB)	Antenna Gain (dBi)	Antenna Gain (dB)	Path Loss (dB)	RSSI (dB)
1	0.1	412	30	14.77121255	3.6	1.45	64.2979443	-48.0767318
2	0.2	412	30	14.77121255	3.6	1.45	70.3185442	-54.0973317
3	0.3	412	30	14.77121255	3.6	1.45	73.8403694	-57.6191569
4	0.4	412	30	14.77121255	3.6	1.45	76.3391441	-60.1179316
5	0.5	412	30	14.77121255	3.6	1.45	78.2773444	-62.0561319
6	0.6	412	30	14.77121255	3.6	1.45	79.8609693	-63.6397568
7	0.7	412	30	14.77121255	3.6	1.45	81.1999051	-64.9786926
8	0.8	412	30	14.77121255	3.6	1.45	82.3597441	-66.1385315
9	0.9	412	30	14.77121255	3.6	1.45	83.3827945	-67.161582
10	1	412	30	14.77121255	3.6	1.45	84.2979443	-68.0767318
11	1.5	412	30	14.77121255	3.6	1.45	87.8197695	-71.598557
12	2	412	30	14.77121255	3.6	1.45	90.3185442	-74.0973317
13	2.5	412	30	14.77121255	3.6	1.45	92.2567445	-76.0355319
14	3	412	30	14.77121255	3.6	1.45	93.8403694	-77.6191569
15	3.5	412	30	14.77121255	3.6	1.45	95.1793052	-78.9580927
16	4	412	30	14.77121255	3.6	1.45	96.3391441	-80.1179316

Table. 5.1: RSSI Vs Distance Analysis.

The graph obtained by the values of Received Signal Strength Intensity (RSSI) Vs Distance is:

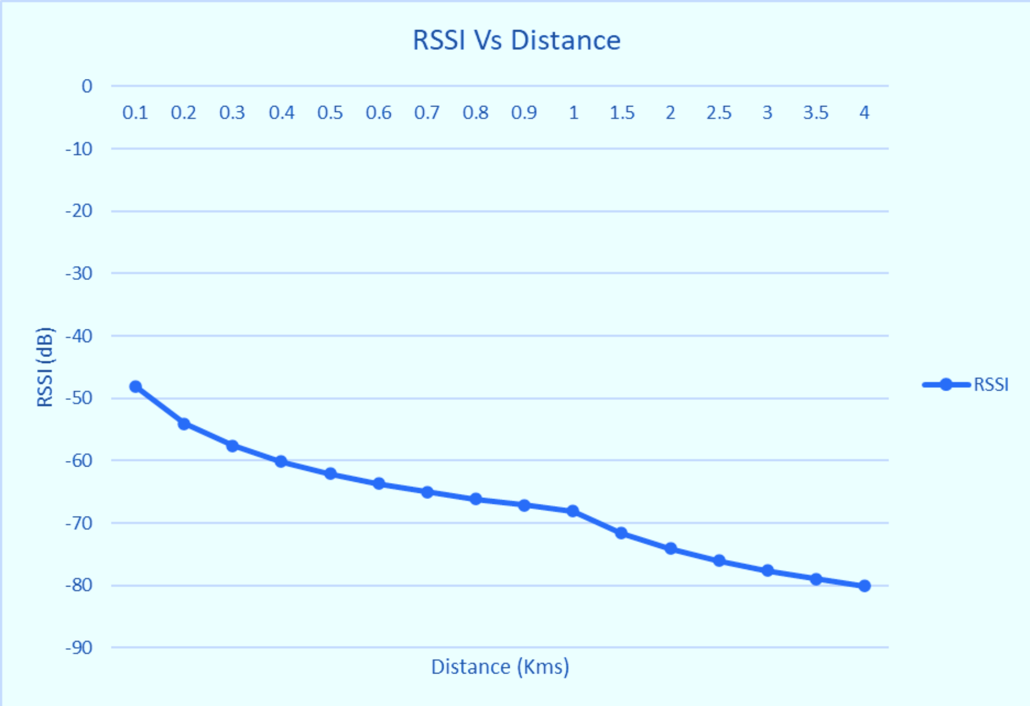

Fig. 5.3: RSSI Vs Distance.

5.3 Path Loss

Path loss, also known as free-space path loss (FSPL) or simply propagation loss, is the reduction in signal strength as a wireless signal travel from a transmitter to a receiver over a certain distance. It is a natural phenomenon that occurs in wireless communication systems and is influenced by various factors.

Path loss is the reduction in power density of an electromagnetic wave as it travels through space. It's also known as path attenuation. Path loss is a key component in the design and analysis of a telecommunication system's link budget. It's usually expressed in decibels (dB).

Path loss is defined as the ratio of transmit power to receive power. In a link budget, this refers to the largest transmit power that the transmitter can send and the smallest receive power at which the receiver can recover the original information. Path loss can occur for a variety of reasons. For example, when communicating from Earth to a geosynchronous satellite, the path loss term is −190 db. Typical values of path–loss exponents range between 1.5 and 5.

Path loss, also known as signal attenuation or free-space loss, refers to the reduction in signal strength as a radio wave travels through the air or any other medium. It is a fundamental concept in wireless communication systems and plays a crucial role in determining the range and coverage of wireless networks.

Several factors contribute to path loss, and understanding them is essential for designing efficient and reliable communication systems. The primary factors influencing path loss include:

- **Distance:** As the distance between the transmitter and receiver increases, the signal strength diminishes. This is a direct consequence of the spreading of the electromagnetic waves over a larger area, leading to a decrease in signal intensity.
- **Frequency of the Signal:** Higher frequency signals experience more significant path loss than lower frequency signals. This is due to the increased absorption and scattering of higher frequency waves by atmospheric gases and objects in their path.
- **Obstacles and Absorption:** Physical obstacles such as buildings, trees, and terrain can block or absorb radio waves, leading to increased path loss. The composition of materials in these obstacles also affects signal absorption, with different materials exhibiting varying degrees of signal attenuation.
- **Atmospheric Conditions:** Weather conditions and atmospheric constituents can impact path loss. For instance, rain, snow, and fog can absorb or scatter radio waves, contributing to signal degradation. In some cases, certain atmospheric conditions can enhance signal propagation.
- **Antenna Characteristics:** The design and characteristics of the antennas used in the communication system influence path loss. Directional antennas may provide a stronger signal in a specific direction but might result in increased path loss in other directions.
- **Multipath Fading:** In a real-world environment, radio waves often take multiple paths to reach the receiver due to reflections, diffraction, and scattering. This can lead to constructive or destructive interference, causing fluctuations in signal strength known as multipath fading.

Usually, the wireless channel is noisy. We can’t have a noise-free communication. We often use the AWGN (Additive White Gaussian Noise) model for interpretation of the channel noise. Noise is assumed to get added to the signal and at the receiver; we have a signal that carries both the data and the noise. It is important that signal level at the receiver is reasonably above the noise floor so that the detector can faithfully detect and decode the signal data.

Apart from noise, losses in the atmosphere can also distort the transmitted signal. Some of the common losses taken into consideration during the design of wireless link budgets include atmospheric absorption loss, scattering loss (includes rain attenuation), reflection loss, diffraction loss and path loss. In this article, we will discuss the path loss and its effects on long distance wireless communications.

5.4 Calculation of Path Loss with the Distance

The Path Loss will be calculated by using a simple formula;

- **Path Loss(dB) = 32 + 20*log (Distance (Km)) + 20*log (Frequency (MHz))**

After calculating the Path Loss for the distances of range (0.1Km to 4Km) we obtain the values of Path Loss in dB as;

Path Loss Vs Distance			
S.No	**Distance (Kms)**	**Frequency (MHz)**	**Path Loss (dB)**
1	0.1	412	64.29794432
2	0.2	412	70.31854423
3	0.3	412	73.84036942
4	0.4	412	76.33914415
5	0.5	412	78.27734441
6	0.6	412	79.86096933
7	0.7	412	81.19990512
8	0.8	412	82.35974406
9	0.9	412	83.38279451
10	1	412	84.29794432
11	1.5	412	87.8197695
12	2	412	90.31854423
13	2.5	412	92.25674449
14	3	412	93.84036942
15	3.5	412	95.17930521
16	4	412	96.33914415

Table. 5.2: Path Loss Vs Distance Analysis.

The graph obtained by the values of Path Loss Vs Distance is:

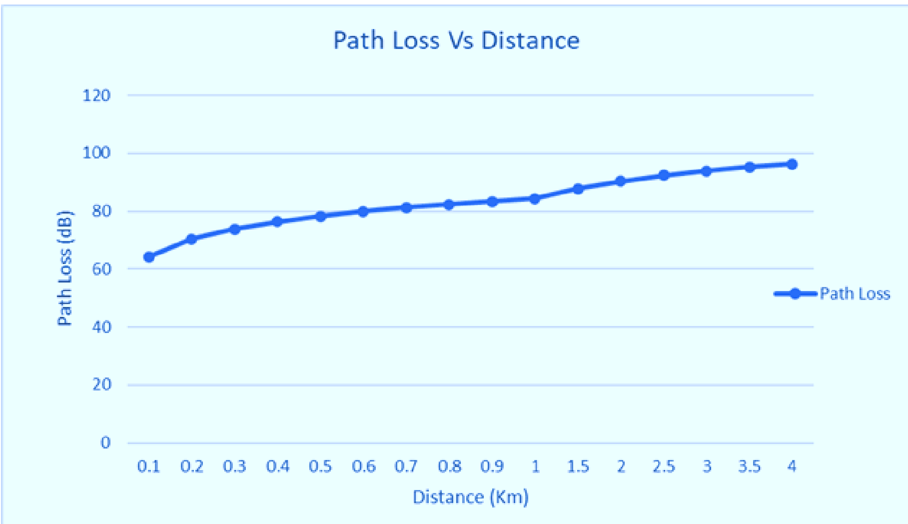

Fig 5.4: Path Loss Vs Distance.

5.5 Long Range Testing of E-Byte E32 433T30D LoRa Module in Line of Sight

A Long-range testing for the E-Byte E32 433T30D LoRa Module has been down to see up to what maximum distance the module can transmit the data. We have observed nearly 3 – 4 Km of distance range in our testings.

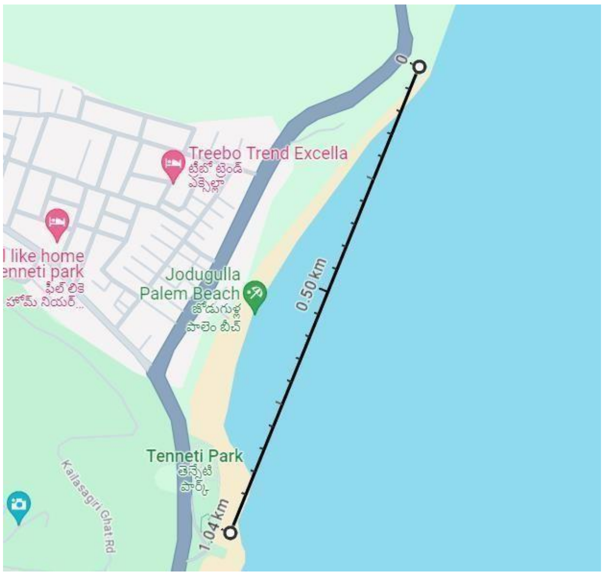

Fig 5.5: Distance range of 1Km.

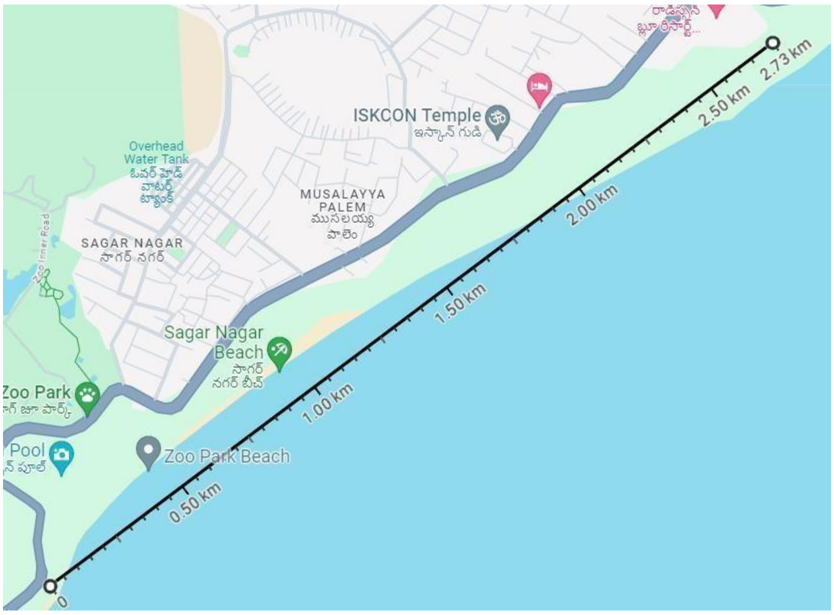

Fig 5.6: Distance range of nearly 3Km.

5.6 High Data Rate Testing of E-Byte E32 433T30D LoRa Module

A High Data Rate is also done on the module in order to find out what will be the maximum data size that the module can send at a time. We have found that the module can send up to 58 Bytes of data at one go.

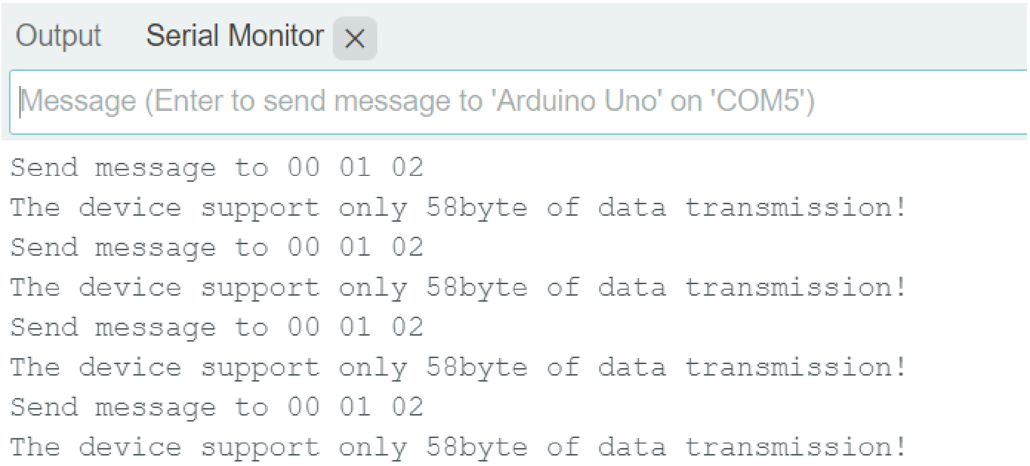

Fig 5.7: High Data Rate Testing.

CONCLUSION

In this project, the LoRaWAN network is created using E-Byte E32433T30D LoRa modules where we have two nodes as senders where they transmit data and a Arduino uno or mega as receiver node. By harnessing the capabilities of the E-Byte E32 433T30D LoRa Modules and configuring them effectively, we were able to demonstrate the impressive range and reliability of LoRaWAN technology, reaching distances of up to 2.5km in non-line-of-sight scenarios. The seamless integration of sensor-equipped nodes with the Arduino gateway further exemplifies the adaptability and versatility of the design, facilitating secure data transmission to multiple cloud platforms.

The project's focus on affordability and accessibility underscores its potential for widespread adoption across diverse sectors, including agriculture, aquaculture, and various industries. By offering a cost-effective alternative to conventional gateways, our solution opens up new possibilities for real-time data monitoring and analysis, enabling informed decision-making and streamlined operations.

Furthermore, the successful implementation of multiple communication channels within the Arduino gateway ensures compatibility with a variety of cloud platforms, enhancing its usability and scalability in different application domains. The robustness and reliability demonstrated in node-to-node and multi-node communication testing validate the efficacy of the solution in facilitating seamless data transmission in complex environments.

Overall, the project's outcomes signify a significant step forward in the democratization of IoT technology, making it more accessible and practical for a wider range of applications. The achievements presented here lay a strong foundation for future advancements in the field of LoRaWAN technology, fostering innovation and progress in the realm of IoT connectivity and data management.

REFERENCES

[1] A. Lavric, A. I. Petrariu, “LoRaWAN communication protocol: The new era of IoT,” 2018 International Conference on Development and Application Systems (DAS), 2018, DOI: 10.1109/DAAS.2018.8396074;

[2] H.-C. Lee, K.-H. Ke, “Monitoring of Large-Area IoT Sensors Using a LoRa Wireless Mesh Network System: Design and Evaluation,” IEEE Transactions on Instrumentation and Measurement, vol. 67, no. 9, pp. 2177-2187, 2018, DOI: 10.1109/TIM.2018.2814082;

[3] K. Tzortzakis, K. Papafotis, P. P. Sotiriadis, “Wireless self-powered environmental monitoring system for smart cities based on LoRa,” 2017 Panhellenic Conference on Electronics and Telecommunications (PACET), 2017, DOI: 10.1109/PACET.2017.8259970;

[4] E. Bingol, M. Kuzlu, M. Pipattanasompom, “A LoRa-based Smart Streetlighting System for Smart Cities,” 2019 7th International Istanbul Smart Grids and Cities Congress and Fair (ICSG), 2019, DOI: 10.1109/SGCF.2019.8782413;

[5] P. Gotthard, T. Jankech, “Low-Cost Car Park Localization Using RSSI in Supervised LoRa Mesh Networks,” 2018 15th Workshop on Positioning, Navigation and Communications (WPNC), 2018, DOI: 10.1109/WPNC.2018.8555792;

[6] Terteil A.A. Ali, V. Choksi, M. B. Potdar, “Precision Agriculture Monitoring System Using Green Internet of Things (G-IoT),” 2018 2nd International Conference on Trends in Electronics and Informatics (ICOEI), 2018, DOI: 10.1109/ICOEI.2018.8553866;

[7] Z. Wenju, L. Shengwei, H. Jiwen, X. Rongtao, L. Hou, “Design and Implementation of Smart Irrigation System Based on LoRa,” 2017 IEEEGlobecom Workshops (GC Wkshps), 2017, DOI: 10.1109/GLOCOMW.2017.8269115;

[8] M. Centenaro, L. Vangelista, A. Zanella, M. Zorzi, “Long-Range Communications in Unlicensed Bands: The Rising Stars in the IoT and Smart City Scenarios,” IEEE Wireless Communications, vol. 23, no. 5, pp. 60-67, 2016, DOI: 10.1109/MWC.2016.7721743;

[9] “LoRaWAN Gateway: Design, Implementation and Testing in Real Environment” by Adrian I. Petrariu, Alexandru Lavric, Eugen Coca

[10] “LoRa Gateway as Internet of Things (IoT) Infrastructure Components on Undip Vocational School” D Y Tadeus, Yuniarto and F Mangkusasmito.

[11] [online] Available: https://github.com/xreef/LoRa E32 Series Library.

[12] SX1278 - E-Byte E32-433T30D LoRa Module datasheet.